

FSC Safety Manager Control Functions

for use with the Honeywell FSC System

Release 680

FS09-600

Implementation
FSC Safety Manager

FSC Safety Manager ***Control Functions***

for use with the Honeywell FSC System
Release 680

FS09-600
11/10

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About This Publication

This publication defines the control functions available for the FSC Safety Manager Module (FSC-SMM) used with the Honeywell FSC system Release 680 and the TotalPlant Solution (TPS) system. It is a reference manual for process engineers, control system engineers and application engineers who design and implement data acquisition and control strategies for TPS systems with Local Control Networks (LCNs).

The user should be familiar with the TPS system control functions described in *System Control Functions* in the *Implementation/Startup & Reconfiguration* - 2 binder before using this publication.

Detailed descriptions of the parameters mentioned in this publication can be found in the *FSC Safety Manager Parameter Reference Dictionary* in this binder, and further background information relative to FSC Safety Manager implementation can be found in *FSC Safety Manager Implementation Guidelines* also located in this binder.

This publication supports TotalPlant Solution (TPS) system network Release 680. TPS is the evolution of TDC 3000^X.

All references in this manual to "FSC Safety Manager" or "FSC Safety Manager Module" pertain only for use with the Honeywell FSC system.
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Acronyms

AI	Analog Input
AM	Application Module
AO	Analog Output
APM	Advanced Process Manager Module
CL	Control Language
CM	Computing Module
DC	Digital Composite
DI	Digital Input
DO	Digital Output
EIP	Event-Initiated Processing
EPROM	Erasable Programmable Read-Only Memory
FLD	Functional Logic Diagram
FSC	Fail Safe Control
FSC-DS	FSC Development System
FSC-SM	FSC Safety Manager
FSC-SMM	FSC Safety Manager Module
I/O	Input/Output
LCN	Local Control Network
LED	Light-Emitting Diode
LM	Logic Manager
NaN	Not a Number
NIM	Network Interface Module
PM	Process Manager
PU	Processing Unit
PV	Process Variable
RAM	Random Access Memory
SM	Safety Manager
SMM	Safety Manager Module
SP	Setpoint
TDC	Total Distributed Control
TPS	TotalPlant Solution
UCN	Universal Control Network
US	Universal Station

If the acronym or abbreviation you are searching for does not appear in this list, refer to the following parameter listing.

Parameters

ALENBST	Alarm Enable Status
ALMOPT	Alarm Option
BADLPFL	Bad Loop Flag
BADPVFL	Bad PV Flag
BADPVTXT	Bad PV State Descriptor
COMMAND	Timer Command
CONTCUT	Contact Cut Out
DISRC(1–2)	Digital Composite Input Connection Source
DLYTIME	Delay Time
DODSTN(1–3)	Digital Composite Output Connection Source
EIPPCODE	Event Initiated Proceeding Point Identifier
EVTOPT	Event Recording Option
FBTIME	Feedback Time
FLLSBA	Least Significant Source Address
FORCE	Force Enable
FORCEFL	Force Flag
GENDESC	Generic Descriptor
HIGHAL	Highest Alarm Detected
HIGHALPR	Highest Level Alarm's Priority
INPTDIR	Input Direction
L	External Output Value
LIBADOPT	Logic Bad-Input Handling Option
LOCALMAN	Local Manual Flag
LOCUTOFF	Low Signal Cut-Off for Flow Inputs
LOENBL(1–12)	Logic Output Enable
MODATTR	Mode Attribute
MODE	Mode
MODEPERM	Mode Permissive
MOMSTATE	Momentary Output States
MOVPVTXT	Moving PV Text Descriptor
NFLAG	Number of Flags
NNLSBA	Least Significant Alias Address
NNUMERIC	Number of Numerics
NODINPTS	Number of Digital Inputs
NODOPTS	Number of Digital Outputs
NONE_OP(1–3)	Value Stored in Output n
NONECONF	Add Optional None State
NOSTATES	Number of Digital States
OFFNRMFL	Off-Normal Alarm Flag
OP	Commanded Output State (Digital Output Point)
OP	Last Digital Output State Requested (Digital Composite Point)
OP	Output in Percent (Analog Output Point)
OPFINAL	Final Output State Read Back from the FSC-SRS (Digital Composite Point)
OPFINAL	Percent Output at the Control Element Analog Output Point)
OPFINHI	Output High Value at Control Element
OPFINLO	Output Low Value at Control Element
OPRATRFL	Operator Mode Attribute Flag
OPTDIR	Analog Output Direct/Reverse Action
PLCADDR	FSC-SRS Alias Address
PNTFORM	Point Form

Continued on next page

Parameters

PTEXECST	Point Execution State
PTINAL	Point in Alarm Indicator
PV	Process Variable Current State (Digital Composite, Digital Input and Flag)
PV	Process Variable Current Value (Analog Input, Numeric and Timer Points)
PVALDB	PV Alarm Dead Band as a Percentage of Full Range
PVAUTO	Current PV State (Digital Composite and Digital Input Points)
PV CALC	Calculated PV
PVCHAR	PV Characterization Option
PVEUHI	PV High Range in Engineering Units
PVEULO	PV Low Range In Engineering Units
PVEXEUHI	PV Extended Engineering Unit Range High
PVEXEULO	PV Extended Engineering Unit Range Low
PVEXHIFL	PV Extended High Range Violation
PVEXLOFL	PV Extended Low Range Violation
PVFL	PV Flag
PVHIFL	PV High Alarm Flag
PVHIPR	PV High Alarm Priority
PVHITP	PV High Alarm Trip Point
PVLOFL	PV Low Alarm Flag
PVLOPR	PV Low Alarm Priority
PVNORMAL	PV Normal State
PVRAW	PV Raw Value (Analog Input Point)
PVRAW	Raw State of Field Contacts (Digital Input Point)
PVSLTSRC	PV Slot Source
PVSRCOPT	PV Source Option
REDTAG	Red Tag State
RV	Remaining Time
SCANRATE	Scan Rate
SLOTNUM	Slot Number
SO	Status Output (Digital Composite and Digital Output Points)
SO	Status Output of Timer (Timer point)
SPSLTSRC	Setpoint Source Address
ST0_OP(1–3)	State 0 Outputs, 1 through 3
ST1_OP(1–3)	State 1 Outputs, 1 through 3
ST2_OP(1–3)	State 2 Outputs, 1 through 3
STATE	Timer State
STATETXT(0–1)	State Descriptor Text
TIMEBASE	Time Base

References

For FSC-SM documentation:

Publication Title	Publication Number	Binder Title	Binder Number
<i>FSC Safety Manager Installation Guide</i>	FS20-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Implementation Guidelines</i>	FS11-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Control Functions</i>	FS09-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Parameter Reference Dictionary</i>	FS09-550	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Configuration Forms</i>	FS88-500	Implementation FSC Safety Manager	TPS 3076
<i>FSC Safety Manager Service Manual</i>	FS13-500	Implementation FSC Safety Manager	TPS 3076

For FSC documentation:

Publication Title	Publication Number
<i>FSC Safety Manual</i>	PM.MAN.8047
<i>FSC Hardware Manual</i>	PM.MAN.8048
<i>FSC Software Manual</i>	PM.MAN.8025

Section 1 – Introduction

1.1 Section Overview

About this section

This section contains an overview of the FSC Safety Manager, its key components and user interfaces. It also provides references to other publications that are useful or necessary during implementation. Topics included in this section are:

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1.1	Section Overview	1
1.2	FSC Safety Manager Overview.....	2
1.3	FSC Safety Manager Module.....	5
1.4	FSC Controller.....	8
1.5	FSC User Station	10
1.6	Universal Station	11

1.2 FSC Safety Manager Overview

Background

The FSC Safety Manager (FSC-SM) is a process-connected device which resides on the Universal Control Network (UCN) of the TotalPlant Solution (TPS) system as a peer to the Process Manager (PM, APM, HPM) and Logic Manager (LM). It provides standard UCN node functionality, and can communicate with the field devices directly or through connected FSCs. The FSC Safety Manager also supports the FSC networking capability. Figure 1-1 illustrates the relationship between FSC-SM and TPS.

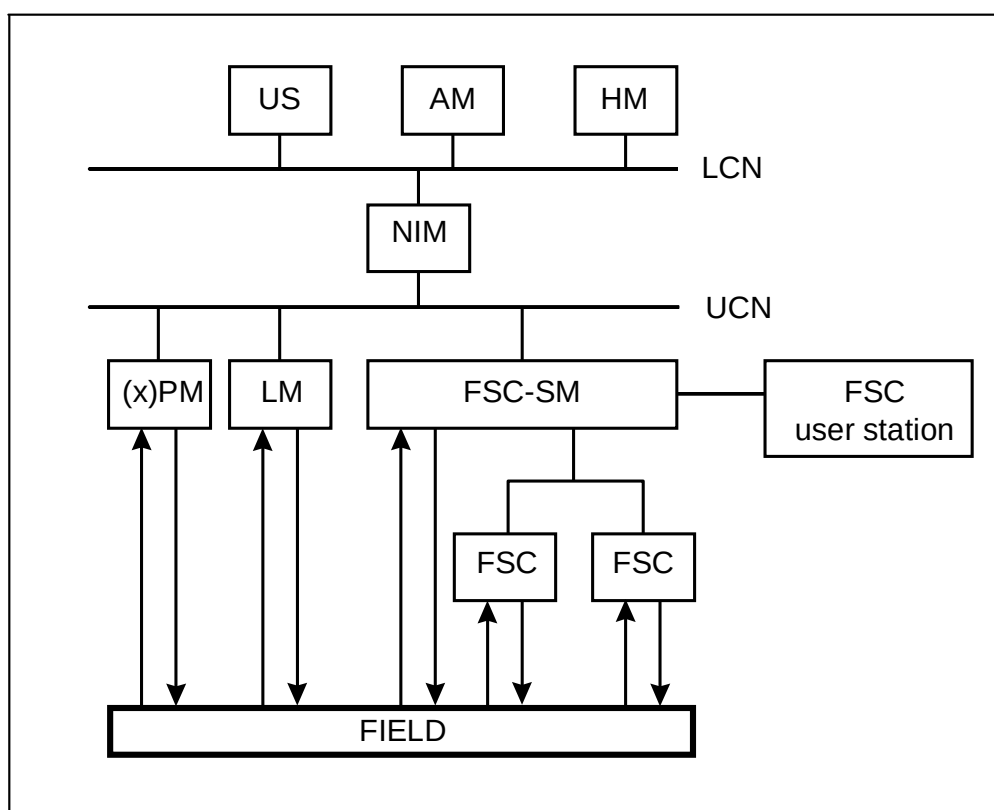


Figure 1-1 – FSC-SM and TPS

Continued on next page

1.2 FSC Safety Manager Overview, Continued

Key components

As illustrated in Figure 1-2, the FSC Safety Manager (FSC-SM) consists of two key components mounted in one or more standard FSC cabinets:

- FSC,
a fail safe control system from Honeywell Safety Management Systems, and
- FSC Safety Manager Module (FSC-SMM),
a Honeywell Safety Management Systems module providing peer-to-peer communications with other process-connected devices residing on the TPS system's Universal Control Network.

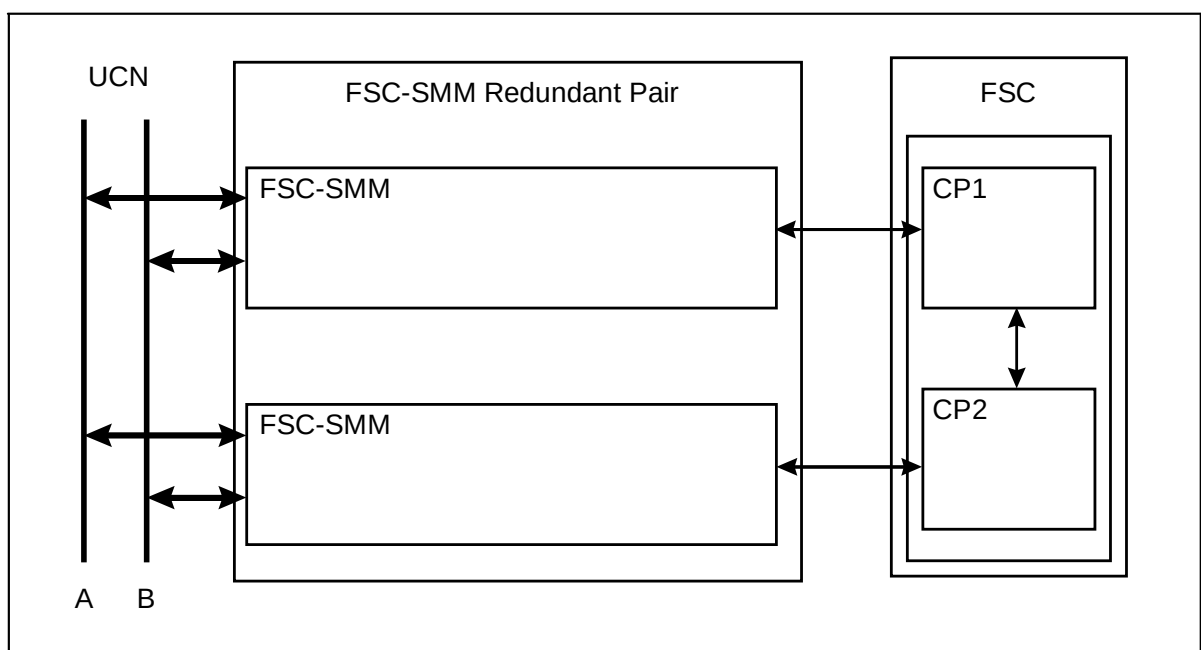


Figure 1-2 – FSC Safety Manager Overview

Available chassis

The FSC Safety Manager uses two types of racks to house the various kinds of available plug-in modules. These chassis are:

- Central Part rack,
housing one or two Central Parts (depending on the FSC configuration), power supplies, communication modules and I/O modules. The configuration of the Central Part rack depends on the FSC configuration rules and the user's choice of modules.
- I/O rack,
housing I/O modules and the module(s) which provide(s) the interface to the Central Part rack.

Continued on next page

1.2 FSC Safety Manager Overview, Continued

Human interfaces

As shown in Figure 1-1, two human interfaces are used in conjunction with the FSC Safety Manager:

- 1) Universal Station (US) and Universal Work Station are two configurations of the TPS-level user interface and can be used to:
 - monitor and control the process, respond to alarms, print reports, and change the equipment status,
 - set up the TPS process database, displays and reports, and load FSC-SMM system software, and
 - diagnose and maintain both control room and process-connected equipment.
 - 2) FSC user station, with the FSC Development System (FSC-DS) software (R42x) or FSC Navigator software (R5xx, R600), which is used to interface with the FSC to:
 - configure FSC hardware and variables,
 - create and load the FSC control program,
 - diagnose FSC system faults,
 - force variables for loop check-out and maintenance of field devices,
 - view FSC control program execution,
 - print engineering documentation,
 - log FSC sequence of events (SOE), and
 - verify the configuration and control program of the FSC component.
-

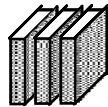
1.3 FSC Safety Manager Module

Introduction

The FSC Safety Manager consists of two key components mounted in one or more standard TPS cabinets:

- FSC,
a fail safe control system from Honeywell Safety Management Systems, and
- FSC Safety Manager Module (FSC-SMM),
a Honeywell optional module to FSC providing peer-to-peer communications with other process-connected devices residing on the TPS system's Universal Control Network.

This subsection provides an overview of the FSC Safety Manager Module component of the FSC Safety Manager.



For more information on FSC redundancy concepts refer to the *FSC Software Manual* and the *FSC Safety Manual*.

Frontplate features

As illustrated in Figure 1-3, the FSC-SMM frontplate includes the following major features:

- Diagnostic and operation LED indicators, which indicate:
 - the result of self-test diagnostics ('STATUS' LED),
 - the UCN transmit status ('Tx' LED),
 - the primary (On) or secondary (Off) ('P' LED),
 - the UCN cable that is currently carrying message traffic ('A' LED / 'B' LED).
- F-style connectors, which provide a link to the redundant Universal Control Network (A and B) via drop cables.

Note:

If the node is not redundant, the 'P' LED will always be off.

Continued on next page

1.3 FSC Safety Manager Module, Continued

Frontplate features,
continued

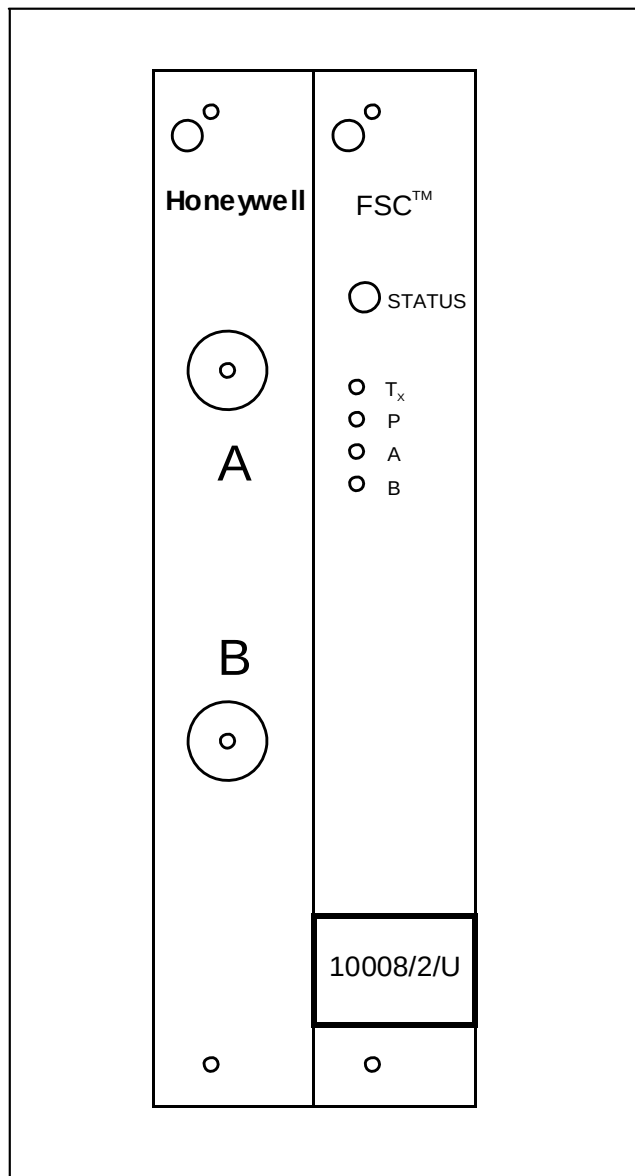


Figure 1-3 – FSC-SMM Frontplate

Continued on next page

1.3 FSC Safety Manager Module, Continued

Function

The FSC Safety Manager Module (FSC-SMM) serves as the communications interface between FSC and the Universal Control Network. The UCN communication channel is redundant. The basic functions of the FSC-SMM are the following (as illustrated in Figure 1-4):

- exchange data between its own tables and the tables of the FSC Control Processor,
- collect and process information from the FSC and its control program:
 - convert the information to TPS data types,
 - perform engineering unit conversions, alarm handling, diagnostic status reporting, peer-to-peer communications, and other UCN functions,
 - pass the converted information to the appropriate devices on the UCN, and
- collect and process information received over the UCN:
 - convert the information to the appropriate formats,
 - pass the converted information to the FSC.

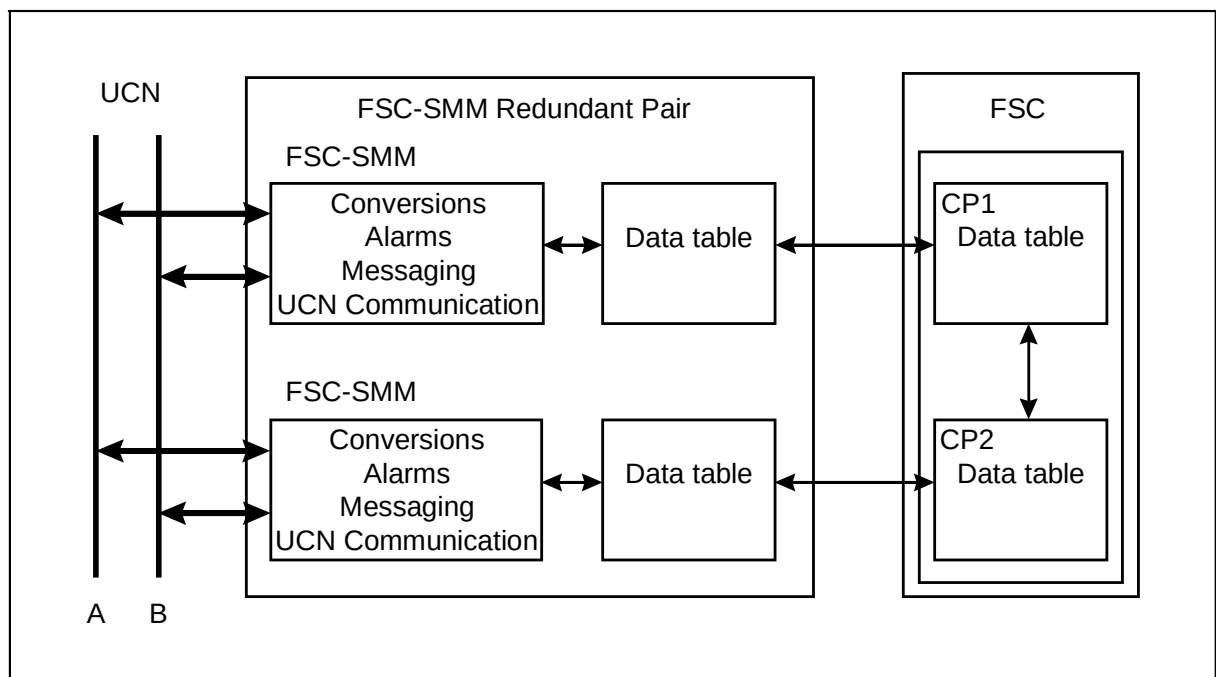


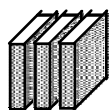
Figure 1-4 – FSC Safety Manager Functions

1.4 FSC Controller

Introduction

The FSC Safety Manager consists of two key components mounted in one or more standard FSC cabinets:

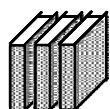
- FSC,
a fail safe control system from Honeywell Safety Management Systems, and
- FSC Safety Manager Module (FSC-SMM),
a Honeywell Safety Management Systems module providing peer-to-peer communications with other process-connected devices residing on the TPS system's Universal Control Network.



For detailed information regarding the concepts and functions described in this section refer to the FSC documentation.

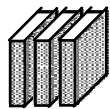
FSC configuration

The configuration of an FSC system depends on the required safety class.



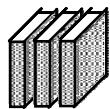
For detailed information regarding the FSC configurations and safety classes refer to the *FSC Safety Manual*.

FSC modules



For detailed information regarding the FSC hardware modules refer to the *FSC Hardware Manual*.

FSC rack configuration

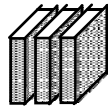


For detailed information regarding the FSC rack configuration refer to the *FSC Software Manual*.

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1.4 FSC Controller, Continued

Networks of FSCs



For detailed information regarding the FSC networks refer to the *FSC Software Manual*.

1.5 FSC User Station

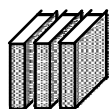
Functions

The FSC component of the FSC Safety Manager is programmed, configured and diagnosed at the FSC user station, which interfaces with the FSC-SM. The FSC user station contains the FSC user software that enables process engineers and maintenance personnel to:

- define the hardware configuration of the FSC,
- define FSC control program variable tagnames,
- develop, test and document the FSC control program,
- diagnose system faults within the FSC, and
- force variables for maintenance purposes.

The FSC user software depends on the FSC release:

- R42x: FSC Development System (FSC-DS), or
- R5xx, R600: FSC Navigator.



For details on the functions, features and use of the FSC user software refer to the *FSC Software Manual* provided with your FSC Safety Manager.

1.6 Universal Station

Background

The Universal Station (US) is the TPS-level user interface to:

- monitor and control the process, respond to alarms, print reports, and change the equipment status,
- set up the TPS process database, displays and reports, and load system software, and
- diagnose and maintain both control room and process-connected equipment.

FSC-SM standard displays

The standard suite of node status displays associated with the FSC Safety Manager and its data points are consistent with similar displays provided for the Logic Manager and Process Manager family of UCN devices.

Display access scenarios

As illustrated in Figure 1-5, access to the US displays associated with FSC Safety Manager status is typical of any other UCN device.

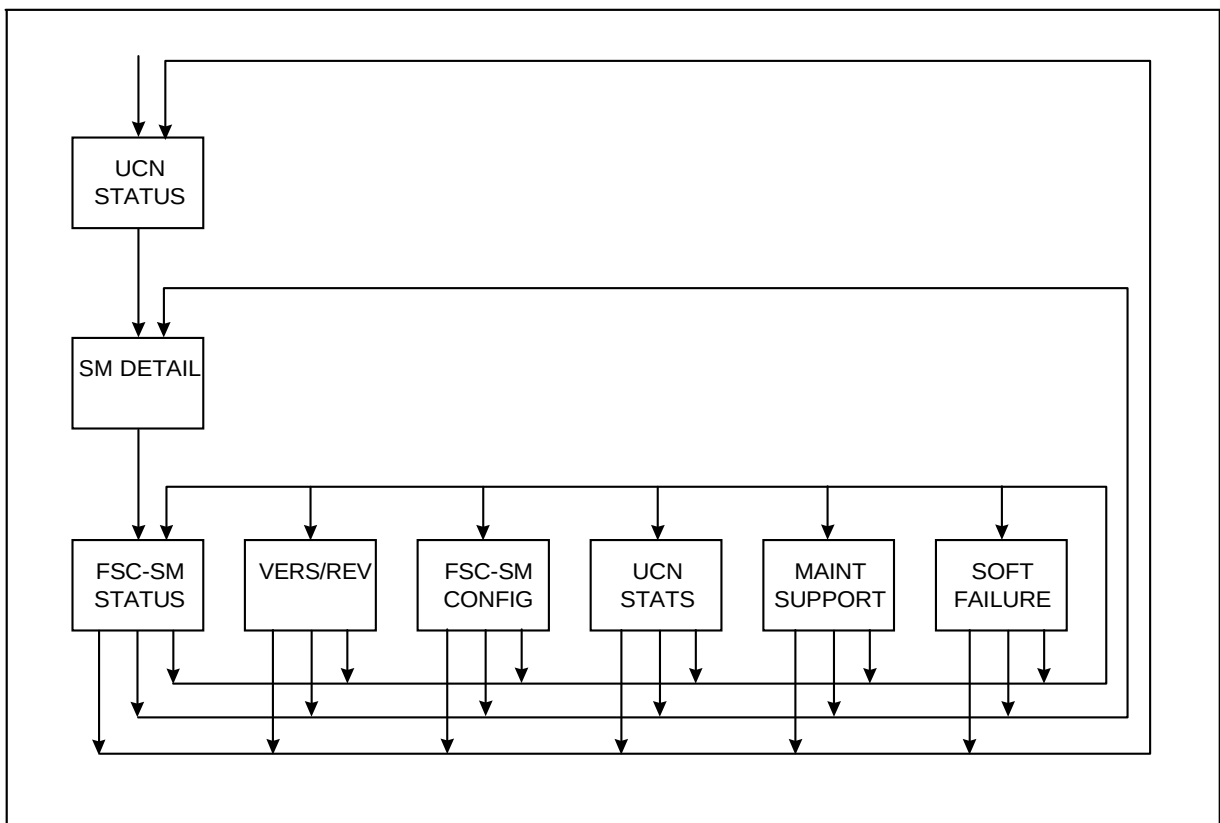


Figure 1-5 – Display Access Scenario

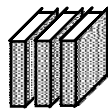
Section 2 – FSC Operation Overview

2.1 Section Overview

About this section

This section provides an overview of the operation of the FSC component of the FSC Safety Manager system. The information presented here provides an overview of the FSC's general operating characteristics. It does not discuss this device's operation in detail. Topics included in this section are:

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For additional and detailed information regarding operating characteristics of the FSC component of the FSC Safety Manager system, refer to the *FSC Software Manual*.

2.2 Operational States

Introduction

The FSC can be in one of five operational states. The FSC state can be retrieved via the FSC-SM Status display, or via the box parameter PLCMODE(1). Figure 2-1 shows the possible transitions from one state to another.

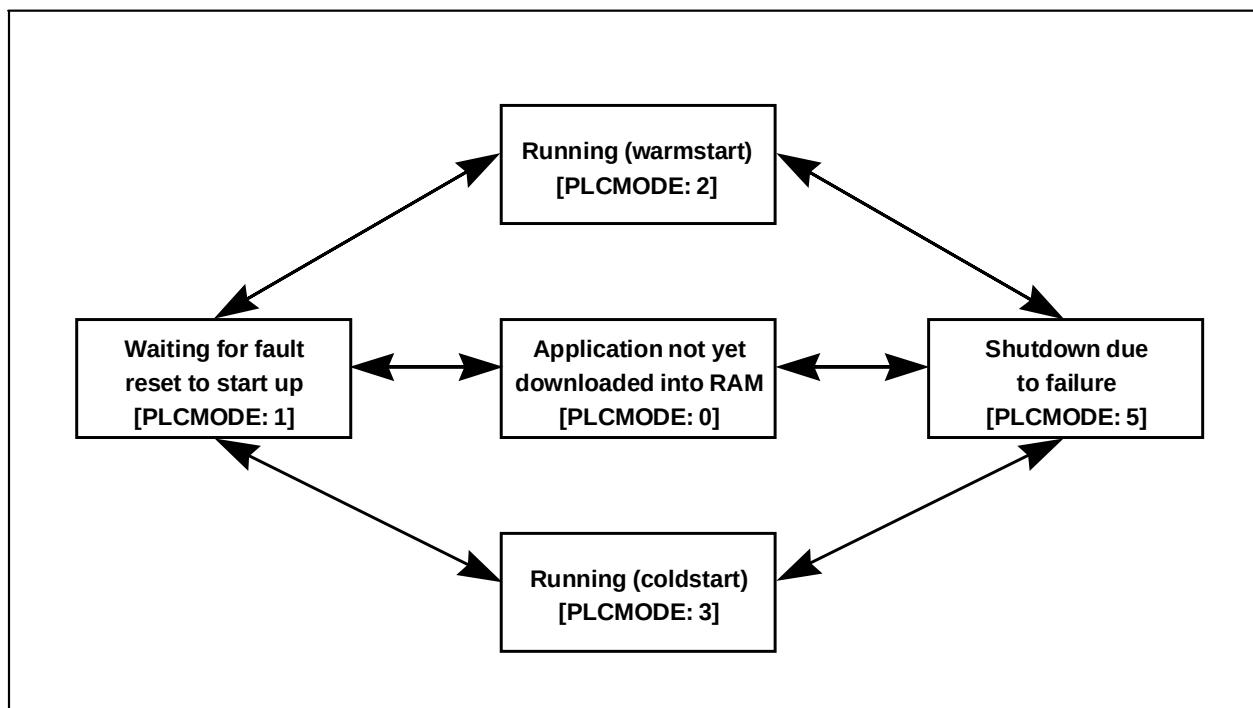


Figure 2-1 – FSC State Transitions

Note:

If PLCMODE = 4, then the system is running in simulation mode.

Description of states

As illustrated in Figure 1-2, the FSC Safety Manager (FSC-SM) consists of two key components mounted in one or more standard FSC cabinets:

- FSC,
a fail safe control system from Honeywell Safety Management Systems, and
- FSC Safety Manager Module (FSC-SMM),
a Honeywell Safety Management Systems module providing peer-to-peer communications with other process-connected devices residing on the Universal Control Network of the TotalPlant Solution (TPS) system.

Continued on next page

2.2 Operational States, Continued

Description of states, continued

The following states can occur:

- *Application not yet downloaded into RAM.* The control program in the FSC can be loaded either into RAM or into EPROM. If the program is loaded into EPROM, this state does not occur. If the program is loaded into RAM, and the program is not yet loaded, the FSC waits for download of its program from the FSC user station.
 - *Waiting for fault reset to start up.* The program has been loaded, and the configuration and program are correct.
 - *Running (warmstart).* The program is running. If the FSC is switched off, and then on again, the program restarts with the last values.
 - *Running (coldstart).* The program is running. If the FSC is switched off, and then on again, the program restarts with the initial power-on values as defined by the user.
 - *Shutdown due to failure.* This state is entered if a failure has been detected which demands shutdown of the FSC.
-

2.3 FSC Control Program

Introduction

The application control program runs in the FSC system and is used for data acquisition, local process/Boolean control algorithms, data transmission to the field, data exchange with other FSCs and data exchange with the FSC user station. The FSC control program is developed using the FSC user software.

ATTENTION The FSC's control program should not be confused with control and data acquisition operations performed at the TPS level. The FSC Safety Manager Module (FSC-SMM) is the interface between the FSC and the TPS system so that data and control operations at the FSC level can be closely integrated with data acquisition and control operations at the TPS level.

FSC user station

With regard to the FSC control program, the FSC user station (with the FSC Development System software or FSC Navigator software) is used to:

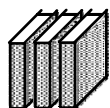
- define the hardware configuration of the FSC,
 - define the FSC control program variable tagnumbers,
 - develop the functional logic diagrams (FLDs).
-

Hardware configuration

The system configuration option of the FSC user software is used to define the physical location and identification of the hardware in the FSC, including the FSC-SMM. This information becomes part of the FSC control program and is downloaded to the FSC. Upon downloading, the location and identification of the system hardware is checked against the FSC control program's configuration definition. Any mismatches are immediately reported.

Tagnumbers

It is not necessary to assign numerical addresses to the FSC control program variables. Instead, the system configuration option and FLD design option of the FSC user software are used to define all program variables using alphanumeric variable names (tagnumbers). A variable is uniquely defined by its type and tagnumber.



For additional and detailed information regarding attributes of variable types refer to the *FSC Software Manual*.

ATTENTION The variables of an FSC control program should not be confused with the Points configured for the FSC-SMM.

2.4 Communication between FSC-SMM and the FSC Control Processor

FSC-SMM – FSC Control Processor memory map

In the FSC-SMM, there are entities called Points. In the FSC Control Processor, there are entities called Variables. If communication between Points and Variables is required, they are linked to each other. This is accomplished by giving them both an address.

In the FSC-SMM, each Point has an address called PLCADDR. In the FSC, each variable can have an address known as a DCS address. A Point and a Variable are linked and their values are exchanged if:

- the Point's and Variable's types match (e.g. AI point and AI variable), and
- the PLCADDR of the Point is equal to the DCS address of the Variable.

FSC-SMM points that require more than one signal to be transferred, have more than one PLC address, with different names. The DCS address of an FSC variable can have any value between 1 and 65535.

FSC-SMM Flag points and FSC-SMM Numeric points have a PLC address defined by their slot number (SLOTNUM) and by the start address (FLLSBA and NNLSBA, respectively). The PLC address of a Flag is $FLLSBA + SLOTNUM - 1$, the PLC address of a Numeric is $NNLSBA + SLOTNUM - 1$. The maximum SLOTNUM that can be assigned to a Flag is NFLAG, the maximum SLOTNUM that can be assigned to a Numeric is NNUMERIC.

ATTENTION The control program within the FSC has primary control over the processing of data which is read into, and written from, its memory. As a consequence, configuration of the FSC-SMM data points must be coordinated with the FSC control program if unexpected results are to be avoided.

Communication mechanism

Both the FSC-SMM and the FSC Control Processor have memory tables where the database of process information is stored and accessed. Figure 2-2 illustrates the communication between these tables.

Continued on next page

2.4 Communication between FSC-SMM and the FSC Control Processor, Continued

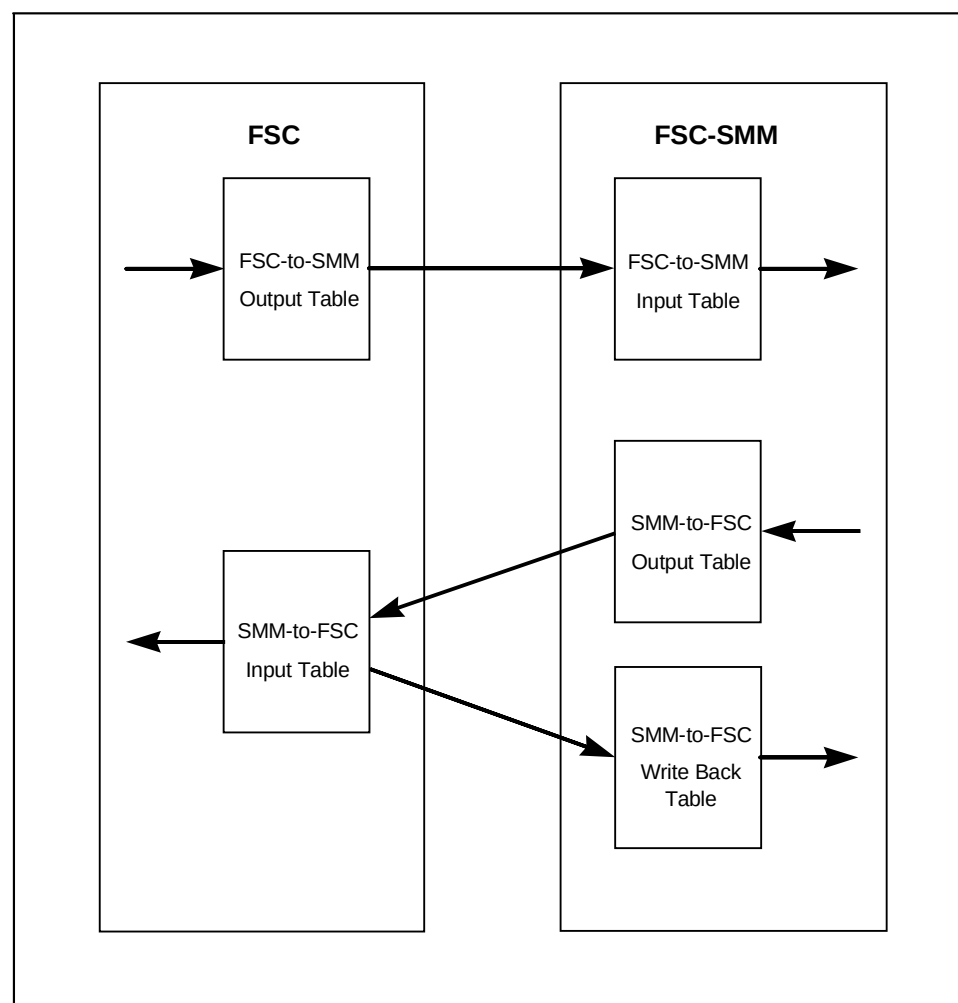


Figure 2-2 – Communication between FSC-SMM and the FSC Control Processor

Communication mechanism, continued

Data that is to be transferred from the FSC to the FSC-SMM is copied by the FSC each FSC cycle from the FSC-to-SMM Output Table to the FSC-to-SMM Input Table. This data can be: field inputs, field outputs, variables in another FSC, or variables from another device which communicates with the FSC.

In addition, during the same FSC cycle, the FSC writes the data back to the SMM-to-FSC Write Back Table. In this way, the FSC-SMM can verify that the data has correctly arrived in the FSC.

Continued on next page

2.4 Communication between FSC-SMM and the FSC Control Processor, Continued

Communication mechanism, continued

ATTENTION

Please note the following:

- FSC-SMM outputs cannot be directly mapped to the FSC field outputs. FSC-SMM outputs are mapped to the SMM-to-FSC Output Table. The control program, running in the FSC, is responsible for reading these FSC-SMM output states from memory, processing them, and actuating any field output. This limitation is essential to maintaining the integrity of the safety shutdown system.
- The effect of Write Back is indicated for AO, DC, and DO points. The value returned from the FSC is stored in a parameter. For the AO point, it is OPFINAL (value from FSC controller, after OPFINHI/OPFINLO range calculation) and OP (OPFINAL after OPTDIR calculation). For the DC point, OPFINAL is the state representation of the field outputs. The actual values of the field outputs is not available in any parameters. For the DO point, it is OP (self-defining enumeration) and SO (Boolean).

Loop status

Each FSC variable has a loop status, which indicates whether the field part of the loop is Healthy or Faulty. The loop status is always Healthy when the input or output channel to which the variable is allocated does not have loop monitoring. In case the channel does have loop monitoring, the loop status of the allocated variable is either Healthy or Faulty. If the loop status is Faulty, the value of the variable cannot be trusted. In that case the BADLPFL parameter of the point corresponding to the variable is set.

Force enable

Each FSC variable has a force enable attribute, which indicates whether that variable can be forced or not. If the force enable attribute is set to Yes in the FSC system, the FORCE parameter of the point corresponding to the variable is set to On.

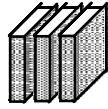
Force flag

Each FSC variable has a force flag, which indicates whether that variable is currently forced or not. If that is the case, the FORCEFL parameter of the point corresponding to the variable is set to On.

2.5 Functional Logic Diagrams

Introduction

The FSC control program is made up of functional logic diagrams (FLDs), which are created using the FSC Development System (FSC-DS) software or FSC Navigator software.



For additional and detailed information regarding functional logic diagrams of the FSC control program, refer to the *FSC Software Manual*.

Section 3 – FSC-SMM Data Point Overview

3.1 Section Overview

About this section

This section contains an overview of the data point types available in the FSC Safety Manager Module (FSC-SMM) component of the FSC Safety Manager (FSC-SM) system. Topics included in this section are:

Subsection	Topic	See Page
3.1	Section Overview	21
3.2	Summary of FSC-SM Data Point Types	22
3.3	Data Point Mix	25
3.4	UCN Device Resource Utilization Display	29
3.5	Execution States	30
3.6	Point Loading and Checkpointing	31
3.7	Point Form	32
3.8	Alarming	33
3.9	Red Tagging	36
3.10	Forcing	37

3.2 Summary of FSC-SM Data Point Types

Introduction

Table 3-1 lists and describes the various TPS data point types available in the FSC Safety Manager Module (FSC-SMM).

Table 3-1 – FSC Safety Manager Module Data Point Types

Point Type	Description
Digital Input (DI)	Digital input from FSC.
Digital Output (DO)	Digital output to FSC. – a memory location for use by the FSC control program.
Digital Composite (DC)	Multiple digital I/O point. – can have 0–2 inputs and 0–3 outputs, and – is typically used for motor or valve control.
Analog Input (AI)	Real value from the FSC — may originate from the FSC as: – raw process input from an analog input module, or – a value supplied by the FSC control program.
Analog Output (AO)	Real value output to FSC. – a memory location for use by the FSC control program.
Logic	Transfers data between the UCN and FSC connections. – supports peer-to-peer communications, and – is known as a Linkage Point in the Logic Manager (LM).
Flag	A two-state point for storing a Boolean value.
Numeric	A point for storing a real value.
Timer	Provides timekeeping functions through an interface with an FSC Timer variable.

Continued on next page

3.2 Summary of FSC-SM Data Point Types, Continued

Data point processing

The order in which data points are processed per scan cycle depends on the scan rate (as defined by the SCANRATE parameter). The SCANRATE parameter can have three values:

AR1DT1: Both analog points and non-analog points (digital, timer, logic, flag) are scanned once per second.

AR2DT2: Both analog points and non-analog points (digital, timer, logic, flag) are scanned twice per second.

AR1DT2: Analog points are scanned once per second and non-analog points (digital, timer, logic, flag) twice per second.

Table 3-2 below shows the order of data point processing in a one-second period.

Table 3-2 – Data point processing per second

SCANRATE parameter			
	AR1DT1	AR2DT2	AR1DT2
1.	first half of analog input (AI) points	all analog input (AI) points	first half of analog input (AI) points
2.	first half of digital input (DI) points	all digital input (DI) points	all digital input (DI) points
3.	first half of digital composite (DC) points	all digital composite (DC) points	all digital composite (DC) points
4.	first half of logic points	all logic points	all logic points
5.	first half of timer points	all timer points	all timer points
6.	first half of flag points (SLOTNUM ≤ 512 only)	all flag points (SLOTNUM ≤ 512 only)	all flag points (SLOTNUM ≤ 512 only)
7.	first half of analog output (AO) points	all analog output (AO) points	first half of analog output (AO) points
8.	first half of digital output (DO) points	all digital output (DO) points	all digital output (DO) points
9.	second half of analog input (AI) points	all analog input (AI) points	second half of analog input (AI) points
10.	second half of digital input (DI) points	all digital input (DI) points	all digital input (DI) points
11.	second half of digital composite (DC) points	all digital composite (DC) points	all digital composite (DC) points
12.	second half of logic points	all logic points	all logic points
13.	second half of timer points	all timer points	all timer points
14.	second half of flag points (SLOTNUM ≤ 512 only)	all flag points (SLOTNUM ≤ 512 only)	all flag points (SLOTNUM ≤ 512 only)
15.	second half of analog output (AO) points	all analog output (AO) points	second half of analog output (AO) points
16.	second half of digital output (DO) points	all digital output (DO) points	all digital output (DO) points

Continued on next page

3.2 Summary of FSC-SM Data Point Types, Continued

Data point processing, continued

ATTENTION Please note the following characteristics regarding data point processing:

- Within one type, the points are executed in ascending SLOTNUM order.
- The order of point processing cannot be configured or altered.
- Do not confuse the FSC-SMM scan with the FSC control program scan, as these two processes are completely independent and run asynchronously to each other.

All UCN and FSC-SMM input information is prefetched prior to their respective processing scan cycles. Output information is poststored after the processing scan cycles. In case of an overrun, all point processing is completed, all end of cycle operations are completed (that includes poststores), and during the next scan, only parameter access is performed until the next cycle comes around (i.e. point process is skipped for one cycle). An excessive number of such scan cycle "slips" will result in an error message.

Point capacity

As part of configuring the FSC-SMM, the required number of points of each type is selected. Each point type has an absolute maximum number per FSC-SMM as shown in Table 3-3. The FSC-SMM point mix must be configured while the FSC-SMM is in the Idle state, and must comply with the performance calculation algorithms as described in the following subsections.

Table 3-3 – Maximum Number of Data Points per Type

Point Type	Maximum Number of Points
Digital Input (DI)	2000
Digital Output (DO)	2000
Digital Composite (DC)	1000
Analog Input (AI)	1000
Analog Output (AO)	1000
Logic Slot	30
Flag	2000
Numeric	1000
Timer	1500

3.3 Data Point Mix

Determining point mix Typically, a mix of different types of points will be configured. To ensure that scanning of all configured points is achieved within the scheduled scan cycle, a per-point weighting system has been established. Each FSC-SMM data point type has been assigned a weight, or point processing unit value. These weights are listed in Table 3-4.

Table 3-4 – Processing Load per Data Point

Point Type	Scan Type	Processing Units per point for 0.5 s scan period	Processing Units per point for 1 s scan period
Digital Input (DI)	digital	1.9	0.95
Digital Output (DO)	digital	1.3	0.65
Digital Composite (DC)	digital	8.1	4.05
Analog Input (AI)	analog	12.5	6.25
Analog Output (AO)	analog	3.1	1.55
Logic Slot	digital	200	100
Flag (SLOTNUM ≤ 512)	digital	0	0
Flag (SLOTNUM > 512)	none	0	0
Numeric	none	0	0
Timer	digital	1.9	0.95

The processing load of a point depends on its scan period. The scan period can be configured independently for digital and analog points. The scan type for each point type is indicated in Table 3-4. Note that the Logic Slot has scan type Digital, even when all inputs of the Logic Slot are analog values.

The scan period can be configured by means of the SCANRATE parameter as shown in Table 3-5.

Table 3-5 – Scan rates

SCANRATE	Analog Scan Frequency	Digital Scan Frequency
AR1DT1	1 per second	1 per second
AR1DT2	1 per second	2 per second
AR2DT2	2 per second	2 per second

Continued on next page

3.3 Data Point Mix, Continued

Validity of point mix

To determine the validity of the intended point mix, follow the procedure described in Table 3-6.

Flag and numeric points use a fixed amount of processing overhead, and therefore are not required to be calculated into the point mix determination.

Table 3-6 – Point Mix Validity Test Procedure

Step	Action	Result
1	Multiply the number of each type of data point in a system by its point processing unit value.	Produces the weighted value of each point type.
2	Add the products produced in Step 1.	Produces the total weighted value of all point types.
3	Test the validity of the point mix.	If the point processing total is less than or equal to 7500, then the total number of data points is valid.

Point mix worksheet

Use Table 3-7 to determine the validity of the intended point mix. Follow the procedure described in Table 3-6 to complete this worksheet.

Continued on next page

3.3 Data Point Mix, Continued

Point mix worksheet, continued

Table 3-7 – Point Mix Worksheet

Point Type	Maximum Allowable Point Count	Processing Unit Value Per Point	X	Number of Points Desired	=	Total Point Processing Units
<i>Digital Input</i>						
0.5 sec. digital scan	2000	1.9	X	_____	=	_____
1.0 sec. digital scan		0.95	X	_____	=	_____
<i>Digital Output</i>						
0.5 sec. digital scan	2000	1.3	X	_____	=	_____
1.0 sec. digital scan		0.65	X	_____	=	_____
<i>Digital Composite</i>						
0.5 sec. digital scan	1000	8.1	X	_____	=	_____
1.0 sec. digital scan		4.05	X	_____	=	_____
<i>Analog Input</i>						
0.5 sec. analog scan	1000	12.5	X	_____	=	_____
1.0 sec. analog scan		6.25	X	_____	=	_____
<i>Analog Output</i>						
0.5-sec. analog scan	1000	3.1	X	_____	=	_____
1.0 sec. analog scan		1.55	X	_____	=	_____
<i>Logic</i>						
0.5 sec. digital scan	30	200	X	_____	=	_____
1.0 sec. digital scan		100	X	_____	=	_____
<i>Timer</i>						
0.5 sec. digital scan	1500	1.9	X	_____	=	_____
1.0 sec. digital scan		0.95	X	_____	=	_____
POINT PROCESSING TOTAL _____						

This page may be reproduced for use as a configuration worksheet.

Continued on next page

3.3 Data Point Mix, Continued

Data point mix considerations

The following considerations apply when determining the data point mix:

- Point processing total must be 7500 or less to be valid.
- The maximum total number of connections between FSC-SMM points and FSC variables is 2000, excluding connections made via flag and numeric points.
- The FSC application cycle time is less than 0.5 second.
- The maximum number of Logic Slots is also limited by the requirement of maximum 50 connections. If all connections of the Logic Slots are used, the maximum number of Logic Slots is four.
- The maximum number of points per Network Interface Module (NIM) is 8000.
- A timer has four FSC addresses: PLCADDR (dual address), PVSLTSRC and SPSLTSRC. If each timer has individual addresses, the maximum number of timers is 500.
- The maximum number of timers with TIMEBASE = Minutes is 446.
- The maximum number of timers with TIMEBASE = Seconds is 748.
- The maximum number of timers with TIMEBASE = 100 Milliseconds is 748.
- The maximum frequency with which other TPS modules can read parameters from the FSC-SMM is 800 per second..
- The maximum frequency with which other TPS modules can write parameters to the FSC-SMM is 100 per second.
- The maximum number of parameter changes that can be communicated from the FSC-SMM to the FSC is 50 per FSC cycle. As a result, there can be a maximum of 50 AO points which receive their value from a continuous controller.
- The maximum rate of continuous events before event overflow occurs is 16 per second.
- The maximum rate of burst events before event overflow occurs is 512 per 10 seconds.

The considerations mentioned above can occur simultaneously. Thus 800 reads per second, 100 writes per second and 50 parameter changes per FSC cycle may take place simultaneously.

3.4 UCN Device Resource Utilization Display

UCN Device Resource Utilization display

With R680, the number of available/consumed PUs can be displayed through the **UCN Device Resource Utilization** display. The display can be accessed through the target UCNDEVUT in the PERFMENU or FSC CONFIG in the Control Configuration Display of an FSC-SM.

The schematic displays the following resource utilization statistics for each slot type:

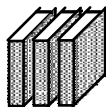
- Total number of PUs consumed by the FSC-SM node.
- Maximum number of PUs available for an FSC-SM node.
- Number of free units in PUs for an FSC-SM node.
- % loaded.

The free units in PUs are indicated in different colors as follows:

- If the free unit is greater than 30%, the value is backlit in green.
- If the free unit is between 20% and 30%, the value is backlit in yellow.
- If the free unit is less than 20%, the value is backlit in red.

In addition to the Resource Utilization statistics, the schematic displays the following:

- Number of points configured per slot type
- Point mix of the selected FSC-SM node
- Available PUs



For detailed information on how to use the UCN Device Resource Utilization display refer to Section 21 of the *Process Operations Manual*. Section 10 of the *Engineer's Reference Manual* explains the details of the displays.

3.5 Execution States

Introduction	A loaded data point is in one of two execution states: Active or Inactive.
Active state	When in the Active state, the point is being processed by the FSC-SMM.
Inactive state	When in the Inactive state, the point is not being processed. Please note the following: • A new or changed point will initially be placed in the Inactive state when being loaded into the FSC-SMM database. • A point must be placed in this state before changes to its configuration can be made. • A point may be set to this state when the FSC application is changed during on-line modification (only applicable for redundant FSC configurations). The point is set to Inactive when the PLC address of the point no longer exists in the new FSC application or the data type is no longer compatible.
Displaying execution states	The Universal Station's Group and Detail displays indicate the state of a point. From the Detail display, the user can also change the point's state by selecting the PTEXECST parameter (Point Execution Status).

3.6 Point Loading and Checkpointing

Point loading and checkpointing

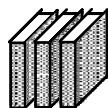
The FSC-SMM database, like those for most other TPS devices, consists of two basic sets of data:

- program instructions (fixed for each release of software), and
- variable configuration data
 - box-related data (applies to entire device), and
 - point data.

Whenever a Safety Manager (or other device) is activated – for the first time or after being powered down for any reason – its FSC-SMM program instructions must be loaded from the History Module or removable media. A simple operating procedure is provided for this purpose. Once loaded, the configuration remains intact unless a failure or power-down occurs.

Device or point data is unique for each device or point. Some of these data types can change or be changed by the user at any time. For this reason, a provision is made to automatically save, or checkpoint, the data on a periodic basis. A manual save can also be requested by the user at any time. If the data is then lost from device memory due to a failure, it can be restored from the checkpointed database.

Parameters associated with the FSC-SMM data points are maintained in tables within the FSC-SMM or within the NIM. The *FSC Safety Manager Parameter Reference Dictionary* indicates for each parameter in which box its value is kept.



For additional details regarding point loading and checkpointing, refer to the *FSC Safety Manager Implementation Guidelines*.

3.7 Point Form

Introduction	Within the TPS system, certain process-related information is grouped together and given a name for purposes of identification. For example, all data pertinent to an input from the process (value, limits, engineering units, etc.) is stored in a database file and assigned a unique point tagname. This enables the system to access all or part of that point's data whenever required.
Point form purposes	To implement many control strategies using Logic Managers (LMs), Process Managers (PMs) or Safety Managers (SMs), the data from several data points must be linked together during the point configuration process. For example, a regulatory control loop might consist of an analog input from the Safety Manager, a sophisticated calculation accomplished by an Application Module (AM), and an output through the Safety Manager. The operator, however, needs only one interface to this control loop, not three. To provide for this situation, the Safety Manager, like the Logic and Process Managers, supplies a configurable parameter called PNTFORM (point form), which defines which points are to be used as the primary operator interface for point data. The PNTFORM parameter provides the user with two choices for point form: Full and Component.
Full point form	Points used for the primary operator interface are configured as having full point form, which includes alarm-related parameters.
Component point form	Points that do not require alarm-related parameters are configured as having component point form. Alarm-related parameters are nonexistent for component points. The component point form should be used for points that: • provide inputs to the full point, • handle the outputs from the full point.

3.8 Alarming

Introduction

As with other UCN-based devices, common alarming functions can be configured for FSC-SMM points. Alarm detection and reporting for a point can be configured only if the full point form is specified (PNTFORM = Full).

Priorities

For each point, a separate alarm priority can be specified for each alarm (for example, PV low alarm can be low priority but PV high alarm can be emergency priority). Alarm priority configuration information is maintained by the NIM. The following alarm priorities are supported for FSC-SMM points:

- Noaction,
 - Journal,
 - Low,
 - High,
 - Emergency,
 - Printer,
 - Jnlprint.
-

Enable status

Alarm enable status is applicable to full point forms, and allows enabling, disabling and inhibition of alarms. This function is accomplished through the ALENBST parameter. The alarm enable status function is resident in the NIM.

Contact cutout

The main purpose of the contact cutout function is to prevent a proliferation of alarms from being reported to the operator. This function can be used to cut out alarms on a point when they are generated because of alarm conditions that have already been detected at other points. Contact cutout is provided for all the point types in the SM and is implemented through the CONTCUT parameter. When the contact cutout state is On, the alarms at the point are cut out; any new alarm detected is not reported on the Alarm Summary display of the Universal Station. The alarms continue to be reported to the journals and to the AM or CM for event-initiated processing (EIP).

3.8 Alarming Continued

Process Alarm Rebroadcast

With R660, acknowledged and unacknowledged process alarms (event-type 21) are rebroadcast if they have not returned to normal (RTN) within the re-alarms interval specified by the \$REALARM parameter. The re-alarm feature is applicable for all types of process alarms that can have an RTN event.

Re-alarms is consistent across all stations in the console in that area. Rebroadcast alarms are also sent to the History Module with a new timestamp, and reported to the RTJ and Process Alarm Journal where they will appear as an alarm.

On the alarm displays, a rebroadcast alarm behaves like a new alarm but has a yellow time stamp.

Rebroadcast alarms appear on the GUS Alarm Summary, and on the following GUS/US native-window displays:

- Area Alarm Summary
- Unit Alarm Summary
- Alarm Annunciator
- Point Detail

The re-alarm feature is implemented using the \$REALARM parameter that has a range of 0 through 9999 minutes. The value specified for this parameter determines the amount of minutes that will elapse before the specified point is re-alarmed. Zero is the default value for the \$REALARM parameter. A 0 value also inactivates the re-alarm feature. The user-configured re-alarm interval is the same value for all process alarms associated with the point.

Point types that use the \$REALARM parameter

The following SM point types contain the \$REALARM parameter:

- Analog Input
- Digital Input
- Digital Composite
- Flag

Modify \$REALARM

The following scenarios describe the change in the alarm handling because of the change in the re-alarms interval while the point is in alarm.

Note:

To prevent unusual or inconsistent re-alarms behavior, do not modify \$REALARM while the point is in alarm.

3.8 Alarming Continued

Scenario 1: Re-alarm interval is modified.

If the re-alarm interval is changed (a non-zero value changed to a different non-zero value) while the point is in alarm, the change is not detected for the current alarm. The change in the interval is detected only for the new alarms subsequently raised for that point. In addition, each alarm for that point is updated with the new interval. Therefore, all alarms of the point are synchronized.

Scenario 2: Re-alarm is cancelled.

If the re-alarm interval is changed (a non-zero value changed to zero) while the point is in alarm, the change is immediately detected and the rebroadcast of the alarms does not occur.

Scenario 3: Re-alarm is initiated.

If the re-alarm interval is changed (zero value changed to non-zero) while the point is in alarm, the change is not detected for the current alarm. The change in the interval is detected only for the new alarms subsequently raised for that point. However, the alarms existing prior to the modification are not rebroadcast.

Multiple alarms on the same point

Multiple alarms on the same point are rebroadcast at the same time. The first alarm occurred for that point determines the schedule for all the subsequent alarms. Therefore, all re-alarms for a point appear together in the alarm displays and journals.

3.9 Red Tagging

Introduction

A point can be red tagged to indicate that it requires maintenance.

Red tagging

This is accomplished by setting the REDTAG parameter to On. Typically, the operator sets the output of the point to a desired safe value before the red tag is put on. Once red tagged,

- the point's mode, mode attribute, external mode switching state, and output cannot be changed, and
- the point cannot be reconfigured or deleted if the red tag is On.

Red tagging is supported for the digital output (DO), analog output (AO), and digital composite (DC) point types. A point must be configured for the full point form in order to be red tagged.

Conditions for red tagging

Before a point can be red tagged, the point's

- MODE must be changed to Man,
- MODATTR must be changed to Operator.

ATTENTION To red tag a digital composite point, the output of the point may not be configured for the momentary state.

3.10 Forcing

Forcing

It is possible to force variables which have been configured with the FORCE ENABLE attribute set to Yes. Forcing is done at the FSC user station (using the FSC Development System software or FSC Navigator software). It is only possible after the FORCE ENABLE key-switch has been set to the Enable position. The value of the variable can then be forced to a user-specified value. The FORCE STATUS attribute of a forced variable is On; for a unforced variable this attribute has the value Off.

If the FSC variable has the force option configured as On, the corresponding FSC-SMM point has the FORCE parameter On. If the FSC variable is configured with the force option Off, the corresponding FSC-SMM point has the FORCE parameter Off. If the value is actually forced, the FORCEFL is On. If the value is actually unforced, the FORCEFL is Off.

Section 4 – Data Point Detailed Description

4.1 Section Overview

About this section This section describes in detail each of the operator-displayable data point types, including their major parameters, and how the points relate to the FSC variable database. Topics included in this section are:

Subsection	Topic	See Page
4.1	Section Overview	39
4.2	Digital Input Point	40
4.3	Digital Output Point	43
4.4	Digital Composite Point.....	44
4.5	Analog Input Point.....	61
4.6	Analog Output Point	66
4.7	Logic Point.....	68
4.8	Flag Point	71
4.9	Numeric Point.....	73
4.10	Timer Point	74

4.2 Digital Input Point

Background

A digital input point converts raw FSC Boolean I/O status into UCN-compatible Boolean formats and performs off-normal alarming. It represents any single FSC Boolean I/O status variable.

The Boolean I/O status variables can be:

- field inputs (signal from field, converted by the FSC, but not changed by the FSC program),
 - field outputs (copy of signal to the field),
 - inputs from the FSC-SMM or other devices (signal from another FSC-SMM, not changed by the FSC program),
 - outputs to the FSC-SMM or other devices (copy of signal to another FSC-SMM),
 - inputs from other FSCs (signal from other FSC, not changed by the FSC program),
 - outputs to other FSCs (copy of signal to another FSC), or
 - local FSC variables.
-

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4.2 Digital Input Point, Continued

Status representation The current state of the PV input is represented on the Universal Station Group and Detail displays as two boxes, as shown in Figure 4-1. The boxes are lit or extinguished depending on the current state of PVRAW and the input direction as configured through the INPTDIR parameter, as shown in Figure 4-1.

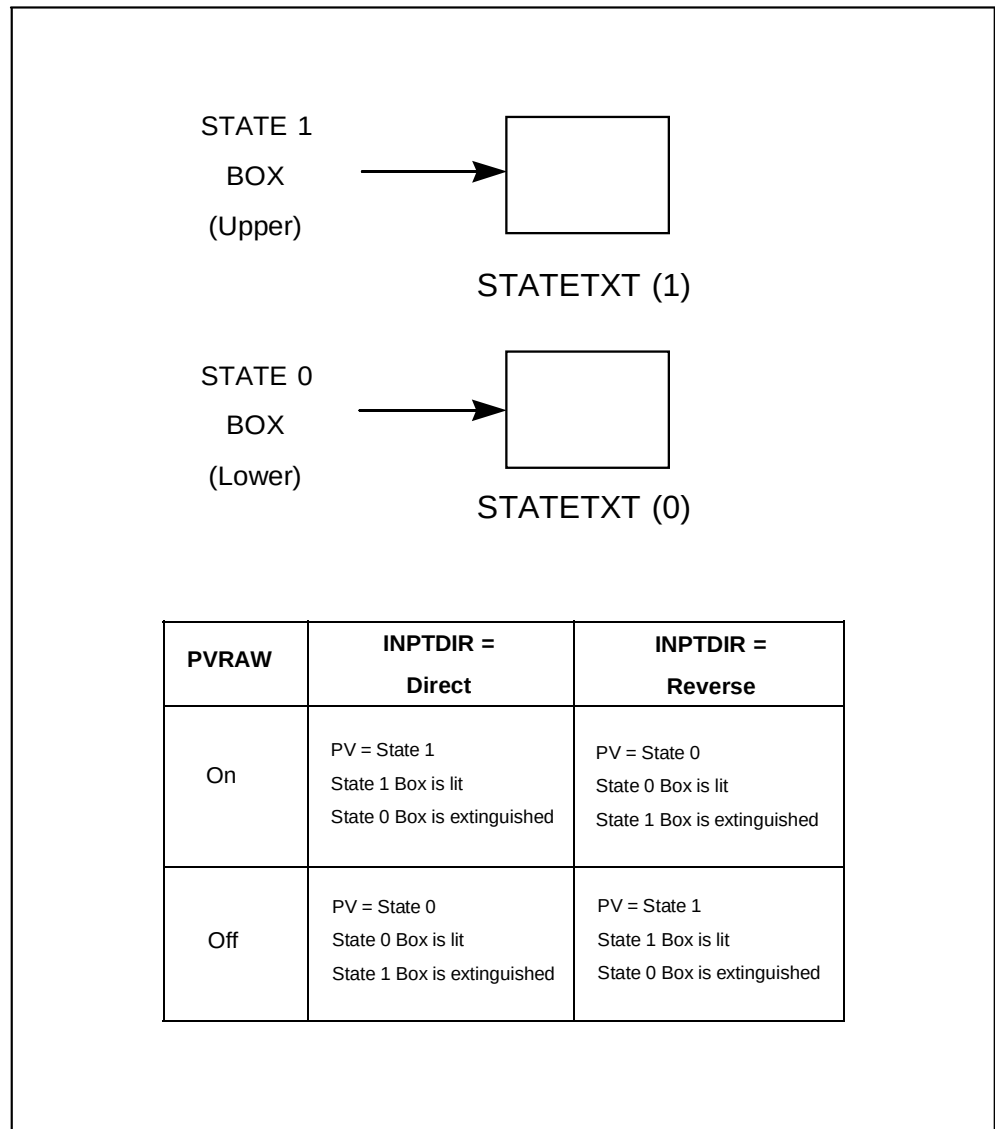


Figure 4-1 – DI Point US Status Representation

Continued on next page

4.2 Digital Input Point, Continued

Point processing

Processing performed on a digital input point is illustrated in Figure 4-2. The PVAUTO value represents the state of the input signal after the direct/reverse conversion is performed. The digital input point can be configured for PV source selection, detection of off-normal alarms, and for reporting any PV state changes to the system.

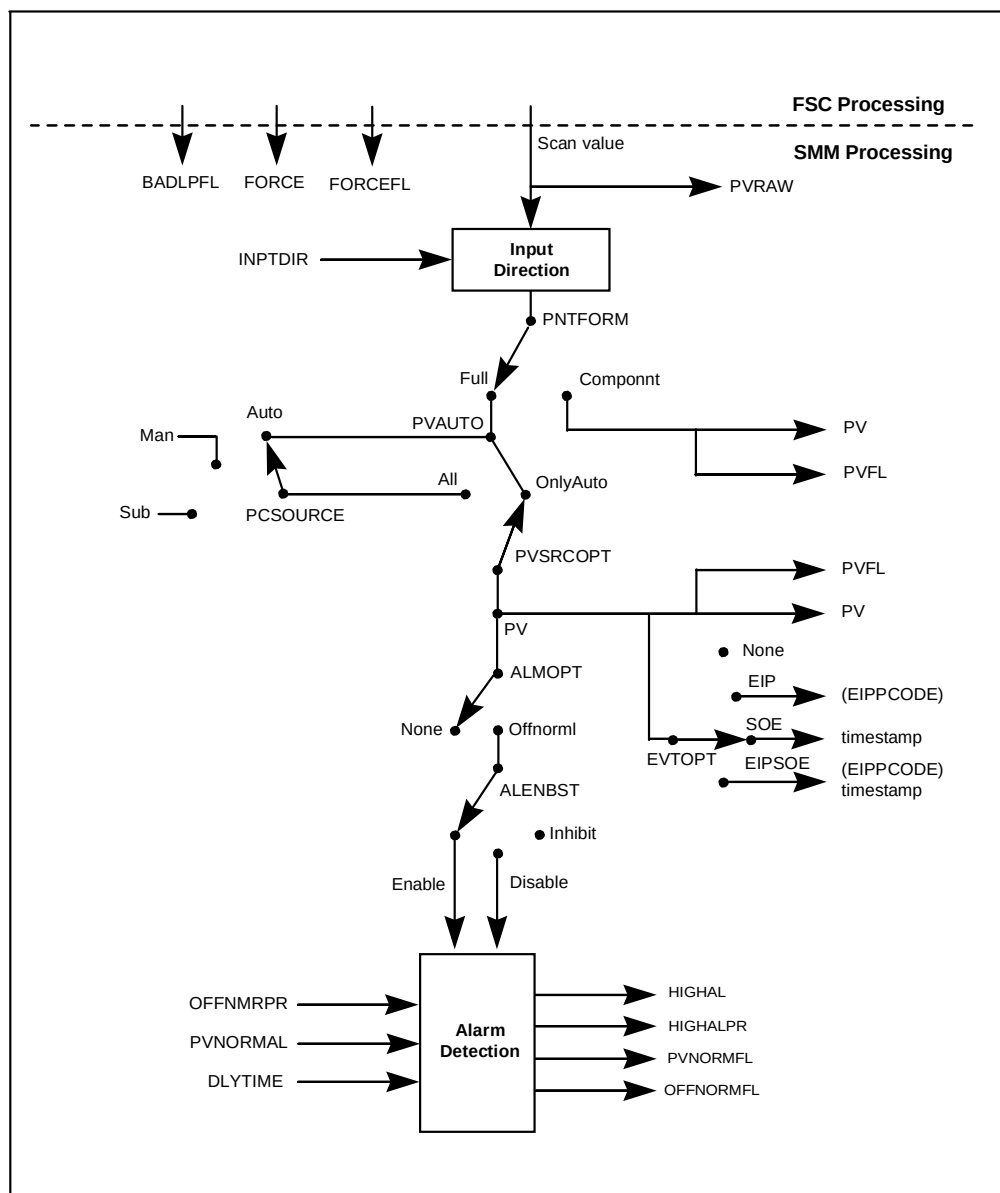


Figure 4-2 – DI Point Processing, Functional Diagram

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4.2 Digital Input Point, Continued

PV source selection

The PV source parameter (PVSOURCE) determines the source of the PV for a digital input point, depending on the PV source option (PVSRCOPT). If the PVSOURCE option is All, the source can be the PV input from the FSC (Auto), the PV state entered by the operator (Man), or it can be supplied by a user program (Sub). If PVSOURCE is Auto, PV tracks PVRAW. If the PV source option (PVSRCOPT) is OnlyAuto, PV tracks PVRAW.

Off-normal alarming

Enabling, Disabling, and Inhibiting Off-Normal Alarms

The ALENBST parameter can be used to enable (permit), disable or inhibit the off-normal alarm. Disabling the alarm still allows the alarm to be listed on the Alarm Summary display. Inhibiting the alarm prevents the current PV state from being compared to the configured normal state.

Off-Normal Alarming and PV Change Reporting

Off-normal alarming can be selected for the digital input point through the ALMOPT parameter. An off-normal alarm is generated when the input PV state is different from the configured normal (desired) state for the point, as specified by the PVNORMAL parameter. The priority of the off-normal alarm is determined through the OFFNRMPR parameter. In addition, any PV state change can be historized as determined by HM volume configuration.

Alarm Delay

When off-normal alarming has been configured and an off-normal alarm is detected, the event is reported to the system. Further off-normal alarms for the same data point are not reported until the time delay (0 to 60 seconds as specified by the DLYTIME parameter) expires. When the time delay expires, the time delay function is disabled and the off-normal alarm for the data point can again be reported.

Continued on next page

4.2 Digital Input Point, Continued

Event reporting

The EVTOPT parameter for the digital input allows the user to:

- optionally specify the tagname (EIPPCODE) of a data point in the system that is to be notified when the PV changes state, and/or
- specify that a time stamp be added to the reported PV state change.

For a status input point, EVTOPT has four possible entries: None, EIP, SOE, and EIPSOE. EIP specifies that the user supply the tagname of the data point in the system that is to receive the PV state change, while SOE specifies that a timestamp is added to the PV state change to establish a sequence of events. EIPSOE is the combination of EIP and SOE.

Suppression of reporting of rapid events

If the EVTOPT parameter is configured as SOE or EIPSOE, reporting of rapid PV changes can be suppressed.

When the PV changes, this is immediately reported. Subsequently nothing is reported during the time specified in PVCHGDLY. After this time has elapsed, the PV is evaluated. When the PV is the same as immediately after the first event, nothing is reported. When the PV is different from the value immediately after the first event, the event immediately following the first one is reported; this report has the timestamp corresponding to the event, not the timestamp corresponding to the report. In both cases, all other possible events are ignored.

4.3 Digital Output Point

Background

The digital output (DO) point value is copied by the FSC from the SMM-to-FSC output table to the SMM-to-FSC input table.

Connections

Depending on the output connection, the DO can be controlled from:

- a DC point output,
- a Logic slot output,
- a CL program, or
- the operator.

Figure 4-3 shows these connections.

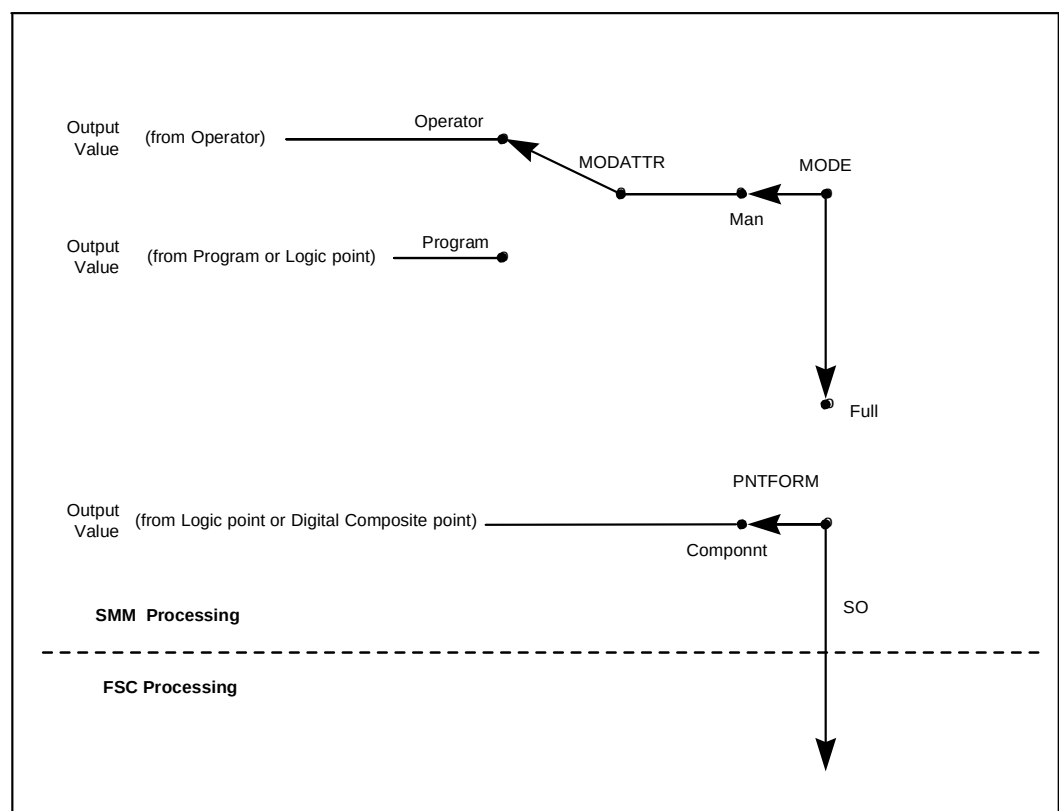


Figure 4-3 – DO Point Processing, Functional Diagram

Output latch functions

The output latch function is obtained by linking a DC point or a Logic slot output connection to the SO parameter of the DO point.

4.4 Digital Composite Point

Background

The digital composite (DC) point is a multi-input/multi-output point which provides an interface to a control algorithm in the FSC. The FSC, in turn, actuates a field device, such as a motor or a valve. The FSC Safety Manager interface allows the operator at a Universal Station to monitor and make inputs to the control scheme through Group, Detail, and Graphic displays.

CAUTION The appropriate algorithm must be programmed into the FSC for the DC point to function. The FSC-SMM uses the FSC variables required to make the DC point's multiple input/output display available to the operator.

Functions performed by the DC point include:

- Commanded states, provided by the operator or a program, pass to the point's outputs for use by the FSC. A Logic slot cannot set the commanded state of a DC point.
- Interlock overrides reflect an override operation built into the FSC control program.
- Because the FSC can alter the state of the outputs to the field device so that it differs from the commanded state, the output of the FSC is read back into the Digital Composite (DC) point so that the operator or the CL program can be aware that the actual output state is not the same as the commanded state.
- FSC variables, local digital inputs, or flags are monitored through feedback inputs. The resulting inputs are checked against the commanded state for Off Normal, Command Disagree, or Uncommanded Change, and alarmed as necessary.
- The status of a local/manual (LOCALMAN) input from the FSC is monitored. No alarming or control is provided. It will prevent writes to the DC point (OP) if the input is on.

Status representation

Figure 4-4 illustrates how DC points are depicted in a Universal Station's Group display. In the display, each state is represented by a separate box. The State 0 box is the middle box (STOP), the State 1 box is the upper box (FORWARD), and State 2 box is the lower box (REVERSE). The State 2 box only appears if the point has been configured for three states (NOSTATES = 3).

Continued on next page

4.4 Digital Composite Point, Continued

Status representation,
continued

ATTENTION

State changes are read back from the FSC, and their indications on the Group or Detail displays may take a few moments after the command to change is issued. This delay can be attributed to:

- any normal lag time required for the actuated field device to change state,
- the scan time for the FSC's logic control program to read the change from the field, and
- the communications time required for transferring data from the FSC, through the FSC-SMM, UCN, NIM, LCN and the US.

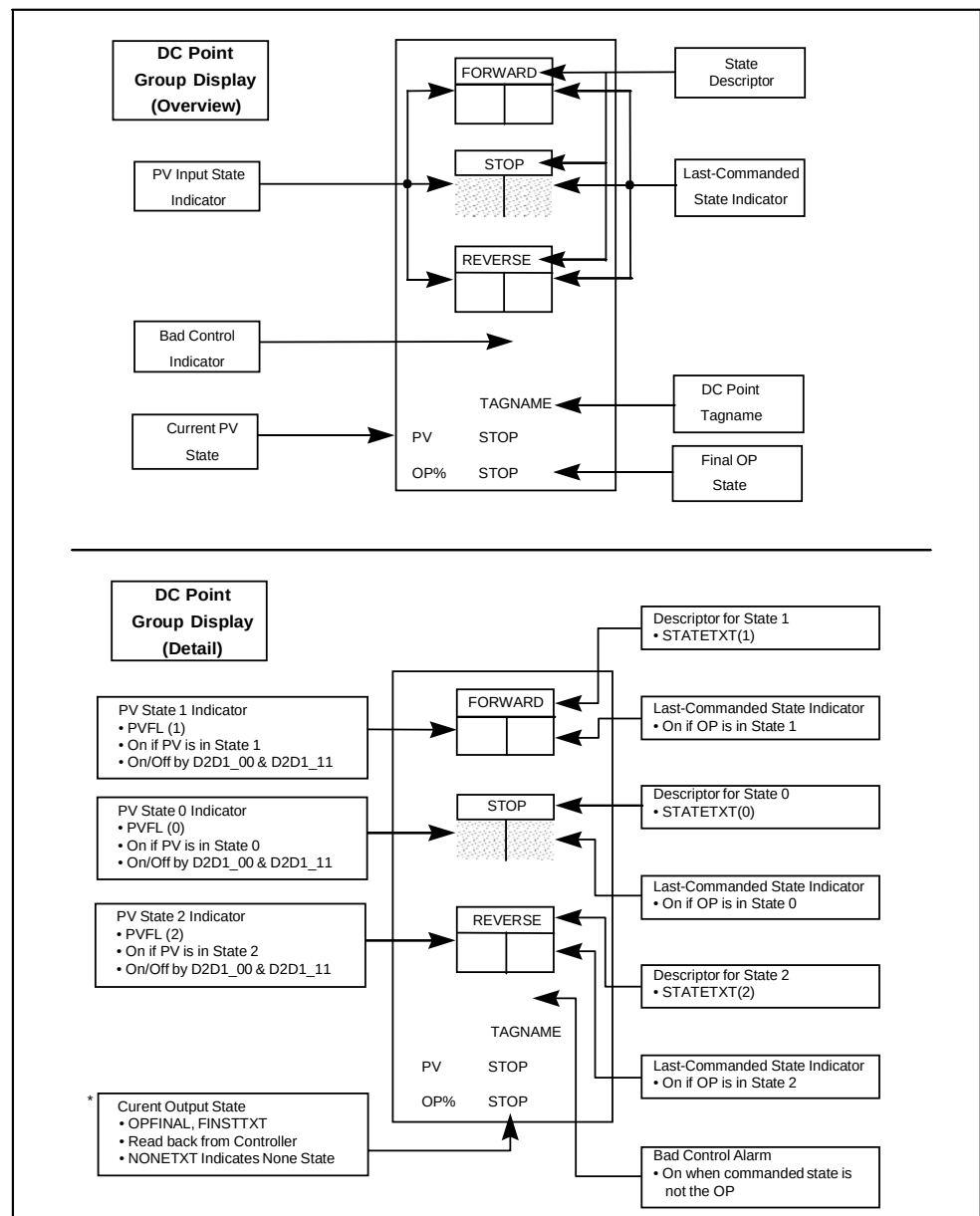


Figure 4-4 – DC Point Group Display

4.4 Digital Composite Point, Continued

Point states

Normal States

DC points can have two or three normal states which allow the data point to interface with devices that have two or three operational states. The normal states of a DC point are defined in Table 4-1.

Table 4-1 – Normal Digital Composite Point State Definitions

State	Definition
1	First active state
0	Inactive (middle) state
2	Second active state

ATTENTION The second active state is only applicable if the entry for the NOSTATES parameter is 3, signifying that the data point has three states.

Momentary States

DC points can be configured as being momentary by using the MOMSTATE parameter. In the momentary state the point acts like a doorbell (state remains active as long as a button is pressed). The momentary states of a DC point are defined in Table 4-2.

Table 4-2 – Momentary Digital Composite Point State Definitions

MOMSTATE	Definition
None	No momentary states.
Mom_1	State 1 is momentary. – When released from this state, it jumps back to State 0.
Mom_0	State 0 is momentary. – When released from this state, it jumps back to State 1. – Mom_0 can be selected only if the parameter NOSTATES = 2.
Mom_2	State 2 is momentary. – When released from this state, it jumps back to State 0. – Mom_2 can be selected only if the parameter NOSTATES = 3.
Mom_1_2	States 1 and 2 are momentary. – When released from any one of these states, it jumps back to State 0. – Mom_1_2 can be selected only if the parameter NOSTATES = 3.

Continued on next page

4.4 Digital Composite Point, Continued

Point states, continued

Moving/Bad States

The digital composite data point has two states which represent the conditions when the current state of the device is "bad" (indeterminate) or the current state is "moving" (from one state to another). The moving/bad state descriptor is located below the State 2 box on the display.

- The Bad State (BADPVTXT parameter) can result when the PV input signals from the process are in an inconsistent state.
 - For example, limit switches on a valve indicating open and closed are both simultaneously on.
- The Moving State (MOVPVTXT) is encountered when the device is in transition from one state to another.
 - For example, a slow moving valve is changing from the open state to the closed state.

The BADPVTXT and MOVPVTXT parameter descriptors are configured once for each FSC-SMM box data point and then are used for all digital composite points in the same FSC-SMM.

None State

The user can create an optional None state for a DC point by configuring On in parameter NONECONF. The None state operates as follows:

NONECONF cannot be configured when NODOPTS = 1, because of the following reason. When there is only one output, it can only be one of two states (one each for the two or three states of the DC point). No other possibilities exist that can be a None (or none of the above) state, as opposed to two outputs (four possibilities) or three outputs (eight possibilities).

If the state read back from the FSC does not match the commanded state in OP or the None state (if configured), then a bad control alarm is not generated.

Users can specify an output state (or a pattern of states if there is more than one output) which does not generate a bad control alarm, even though it differs from the commanded state.

OPFINAL

OPFINAL represents the current state of the actual outputs as described in Table 4-3. The output values are first written by the FSC and then compared to the STx_OPy parameters for a match.

Continued on next page

4.4 Digital Composite Point, Continued

Point states, continued

Table 4-3 – Digital Composite Point OPFINAL operation

If the field outputs match the ...	Then OPFINAL ...
STx_OPy for the current state in OPFINAL (assuming OPFINAL is not Bad)	does not change
ST0_OPy	is set to State 0
ST1_OPy	is set to State 1
ST2_OPy	is set to State 2
NONE_OPy and NONECONF is On	is set to None state
otherwise	is set to Bad state

The outputs are first compared to that of the current state, because more than one state may have the same output values, and OPFINAL should not change to a different state that happens to have the same output value configuration.

Mode attributes

DC points support only the manual (Man) mode with mode attributes (specified by parameter MODATTR) of Operator and Program. Table 4-4 defines how the commanded state is provided.

Table 4-4 – Digital Composite Point Mode Attribute Operation

If MODATTR is	Then ...
Operator	only the Universal Station operator can provide the commanded output state.
Program	only user programs (continuous or discontinuous) can provide the commanded output state.

The operator mode attribute flag (OPRATRFL) is also provided for potential use by the interlocking logic.

The MODEPERM parameter is provided to optionally prevent the operator (as opposed to the supervisor or engineer) at the Universal Station from changing the mode attribute.

Interlocks

Two kinds of interlocks, permissives and overrides, can be programmed in the FSC control program for controlling the DC points associated real-world FSC output(s). When they are so programmed, their positions (On, Off) are communicated to the FSC-SMM. Table 4-5 defines to what state the commanded state is forced.

Continued on next page

4.4 Digital Composite Point, Continued

Interlocks, continued

Permissive interlocks

The permissive interlock parameters P0, P1 and P2 determine whether the operator and user programs are allowed to change the output of a DC point to a specific state. A permissive interlock can be provided for each of the three states. Permissive interlocks themselves never cause the OP to change.

For the commanded output to be changed to the desired state, the corresponding permissive interlock parameter must be set to On. The permissive interlock parameters P0, P1, and P2 are normally set to On, thereby allowing permission for all states. These parameters must be individually set to Off by the FSC control program to prevent access to the corresponding commanded output state.

Override interlocks

The override interlock parameters I0, I1 and I2 force the commanded output (OP) to a specific state regardless of the condition of the permissive interlocks. The operator and user programs cannot change the output state when any override interlock is On. An override interlock can be provided for each of the three output states. Note that the actual outputs to the process are supplied only by command from the FSC control program.

Table 4-5 – Digital Composite Point Override Interlock Operation

If override interlock ...	Then ...
I0 is set to On by the FSC control program	the commanded output is forced to State 0 (regardless of the condition of parameters I1 and I2)
I0 is set to Off and I1 is set to On by the FSC control program	the commanded output is forced to State 1 (regardless of the condition of parameter I2)
I0 and I1 are set to Off and I2 is set to On by the FSC control program	the commanded output is forced to State 2
I0, I1, and I2 are all set to Off by the FSC control program	the last value of the commanded output is maintained until changed by the operator, program or another override interlock

Continued on next page

4.4 Digital Composite Point, Continued

Interlocks, continued

Override interlock parameters I0, I1 and I2 are normally set to Off, disabling all the override interlocks. They must be set to On (by the FSC control program) to force OP only (not the field outputs) to go to any specific state.

ATTENTION

- On the FSC Safety Manager, the override interlocks only change OP, not the field outputs. (The relation between OP and field outputs is created in the FSC control program.)
- A Logic Slot cannot set the values of the interlock parameters Px and Ix, even when the parameters PISRC(x) and OISRC(x) have the value 0 (= not configured). If PISRC(x) has the value 0, the corresponding permissive interlock Px is On. If OISRC(x) has the value 0, the corresponding override interlock Ix is Off.

Digital inputs

When configuring digital inputs of the DC data point, the user can specify

- the input connections,
- PV states,
- PV source and options,
- alarming, and
- change-of-state events.

The input portion of a DC point can be configured only if the user has entered 1 or 2 for the NODINPTS parameter (Number of Digital Inputs).

Current input state (PV)

The DC point allows the user the flexibility to assign the states of the PV for each possible combination of digital inputs, so that the states correspond to the different applications in which this point type can be used. The PV parameter represents the current state of the interfaced device and is derived from inputs D1 and D2 that can be feedback signals from the process. Separate parameters are used to configure a single-input point and a dual-input point.

Continued on next page

4.4 Digital Composite Point, Continued

Digital inputs,
continued

Table 4-6 describes the characteristics associated with single-point and dual-point input parameters.

Table 4-6 – DC Point Single and Dual Digital Input Parameter Characteristics

Point Use	Comment	Assigned Conditions
Single Input	• One input parameter (D1) • Two possible PV states: – Pvstate0 – Pvstate1 • Assigned to either one of two input conditions	• D1 is On; parameter D1_1 • D1 is Off; parameter D1_0
Dual Input	• Two input parameters (D2 and D1) • Four possible combinations of input conditions • User can assign any one of five PV states: – Pvstate1 – Pvstate0 – Pvstate2 – Movpv – Badpv	• D2 is Off, D1 is Off; parameter D2D1_00 • D2 is Off, D1 is On; parameter D2D1_01 • D2 is On, D1 is Off; parameter D2D1_10 • D2 is On, D1 is On; parameter D2D1_11

Single Point: Only Pvstate0 or Pvstate1 need to be assigned to parameter D1_1; the system automatically assigns the remaining PV state to parameter D1_0.

Dual Point: Pvstate1, Pvstate0 and Pvstate2 cause the PV indicator to be lit in the respective state box on the Group display when the assigned D2D1 input conditions are satisfied. The Movpv and Badpv states cause the respective MOVPVTEXT or BADPVTEXT descriptor to appear below the state boxes on the Group display.

Continued on next page

4.4 Digital Composite Point, Continued

Digital inputs, continued

Inputs to a digital composite point are usually the PVs from digital input points. When the Digital Input point is configured as component point, its input direction is direct, and the direct/reverse action can be configured by assigning the appropriate PV state parameters of the Digital Component point. If the Digital Input point is configured as full point, its input direction can be configured with the INPTDIR parameter, and the direct/reverse action can also be configured by assigning the appropriate PV state parameters of the Digital Component point.

PV source

The PV source parameter (PVSOURCE) determines the source of the current PV state for the digital input portion of the digital composite data point. Table 4-7 lists the possible sources of the current PV state.

Table 4-7 – Current PV Source States

Source	Description
Man (manual)	Current PV state is provided by the operator from the Universal Station
Auto (automatic)	Current PV state is derived from Input 1 (D1) and Input 2 (D2)
Track	Current PV state is the commanded output state
Sub (substituted)	Current PV state is provided by a user program

By configuration, the user can specify the PV sources to be used for this data point. Parameter PVSRCOPT allows the user to select the PV source as being only Auto, or to select all the PV sources in the above listing as allowable sources of the PV.

Input Connections

The inputs to a DC point are specified by the user through digital input-source parameters DISRC(1)-DISRC(2). The status of Input 1 is represented by parameter D1; input 2 is represented by parameter D2. Input 2 can be configured only when the entry for the number-of-digital-inputs parameter (NODINPTS) is 2. These inputs are designated as Input 1 and Input 2, and can be obtained from:

- FSC connection,
- SO parameter of a digital output point within the same FSC-SMM,
- PVFL parameter of a local digital input,
- PVFL parameter of a local flag point, and
- FSC-SMM box flag PV.

The input sources must be in the same FSC-SMM box as the DC point.

Continued on next page

4.4 Digital Composite Point, Continued

Digital inputs,
continued

Table 4-8 describes the procedures for assigning DC input point connections.

Table 4-8 – Digital Composite Point Input Connections

To assign ...	Then ...
an FSC connection	enter the following information for the respective DISRC(n) input connection parameter ... !LCxxxx where !LCxxxx is the address of the variable as programmed in the FSC.
a local DI point	enter the following information for the respective DISRC(n) output connection parameter ... Tagname.PVFL where the tagname is the name assigned to the logic slot providing the output and PVFL signifies the PV flag.
a local DO point	enter the following information for the respective DODSTN(n) output connection parameter ... Tagname.SO where the tagname is the name assigned to the data point through the NAME parameter and SO signifies the writeable status command of the digital output point.
an FSC-SMM box flag in the same FSC-SMM	enter the following information for the respective DISRC(n) output connection parameter ... Tagname.PVFL or !BOX.FL(nnnn) or \$NMhhBxx.FL(nnnn) where the Tagname is the name assigned to the box PV flag, PVFL signifies the PV flag, !BOX specifies the same SM device where the DC point resides, nnnn is the flag's slot number, hh is the SM's UCN network number, and xx is the SM's UCN address.

Continued on next page

4.4 Digital Composite Point, Continued

Digital inputs,
continued

DC Input Point Processing

Figure 4-5 provides a functional diagram illustrating the processing of a DC input point.

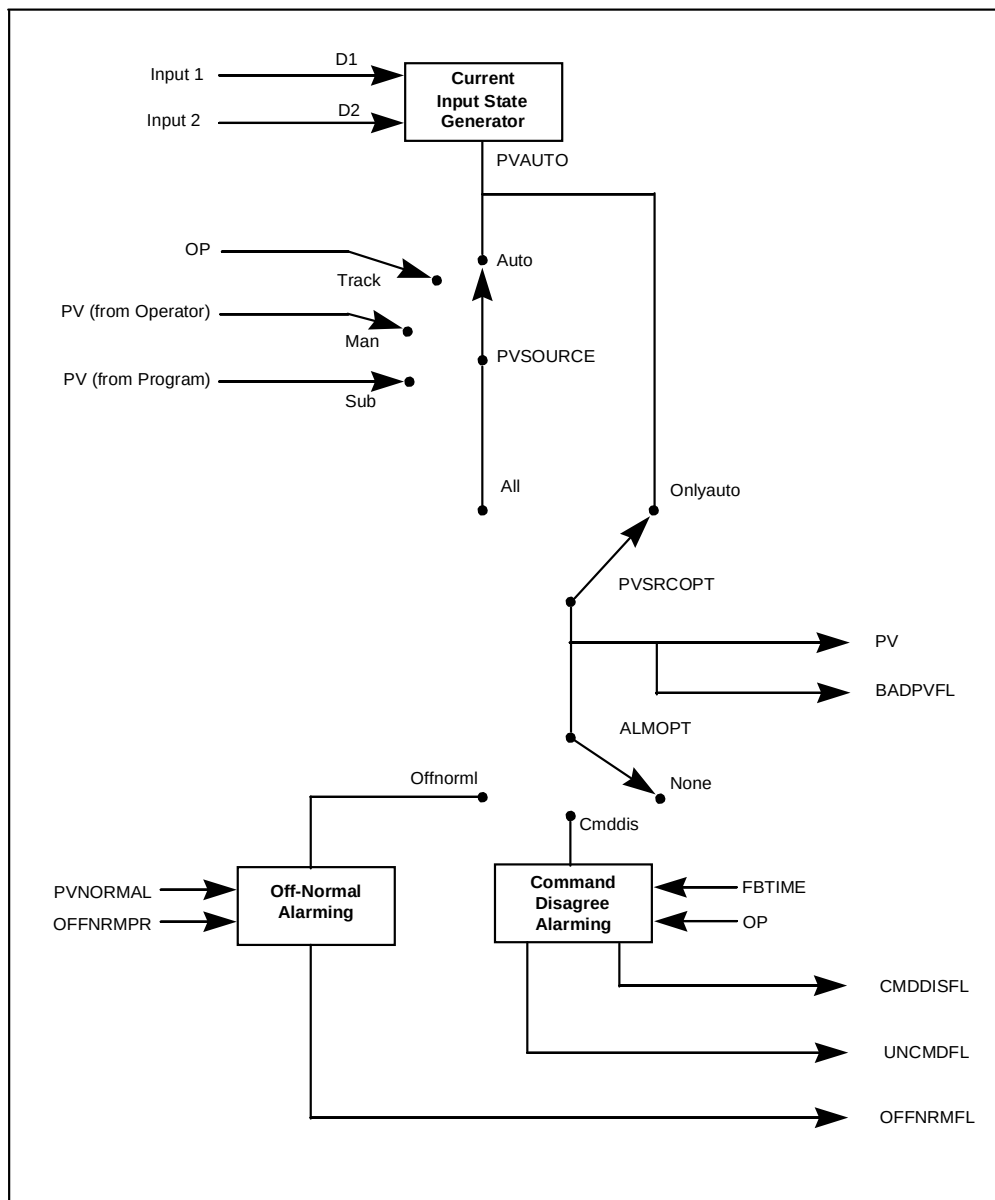


Figure 4-5 – DC Input Point Processing, Functional Diagram

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4.4 Digital Composite Point, Continued

Digital outputs

Digital outputs for the digital composite (DC) point can be configured as either latched or momentary. Momentary outputs use the MOMSTATE parameter.

Commanded state (OP)

The command to go to a specific state normally results in the output to the field device going to the commanded state. (The FSC can alter output state to the field device.) The OP parameter in the DC point contains the descriptor for the commanded state. The descriptor is configured in parameters STATETXT(0) through STATETXT(2). The OP parameter is available for configuration only if the number of digital output connections (NODOPTS) is configured to be greater than 0.

Up to nine Boolean parameters of the form STx_OPy (where x = 0, 1, or 2 for the state number and y = 1, 2, or 3 for the output number) allow the user to specify the normal state values that are to be stored by the output connections. For each of the 3 commanded states per output connection, the user must define the value (On or Off) of the state.

For the None state, the output values for the 3 possible outputs are configured in parameters NONE_OP1, NONE_OP2 and NONE_OP3.

SO(0), SO(1) and SO(2) represent the State Output of the DC point. They are another way of representing OP, but with Boolean values instead of a self-defining enumeration. SO(0) will be On (the others Off) if OP is State 0, SO(1) will be On (the others Off) if OP is State 1, and SO(2) will be On (the others Off) if OP is State 2. Also, for example, writing to SO(1) to On is the same as writing OP to State 1.

Output connections

The destinations of the outputs (and the output types) from a DC point are specified by the user through the DODSTN(1) - DODSTN(3) parameters. Destinations that can be specified are:

- FSC connection,
- SO parameter of a digital output point within the same FSC-SMM,
- PVFL parameter of a local flag point, and
- FSC-SMM box flag.

The output destinations must be in the same FSC-SMM box as the DC point.

Continued on next page

4.4 Digital Composite Point, Continued

Digital outputs,
continued

Table 4-9 describes the possibilities for assigning DC output point connections.

Table 4-9 – Digital Composite Point Output Connections

To assign DC output point to ...	Then ...
an FSC connection	enter the following information for the respective DODSTN(n) output connection parameter ... !LCxxxxx where !LCxxxxx is the address of the variable as programmed in the FSC.
a local digital output point	enter the following information for the respective DODSTN(n) output connection parameter ... Tagname.SO where the tagname is the name assigned to the data point through the NAME parameter and SO signifies the writeable status command of the digital output point.
a local flag point	enter the following information for the respective DODSTN(n) output connection parameter ... Tagname.PVFL where the tagname is the name assigned to the data point through the NAME parameter and PVFL signifies the PV flag.
an FSC-SMM box flag in the same FSC-SMM	enter the following information for the respective DODSTN(n) output connection parameter ... !BOX.FL(nnnn) or \$NMhhBxx.FL(nnnn) where !BOX specifies the same SM device where the DC point resides, nnnn is the flag's slot number, hh is the SM's UCN network number, and xx is the SM's UCN address.

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4.4 Digital Composite Point, Continued

Digital outputs,
continued

DC Output Point Processing

Figure 4-6 provides a functional diagram of the processing of a DC output point.

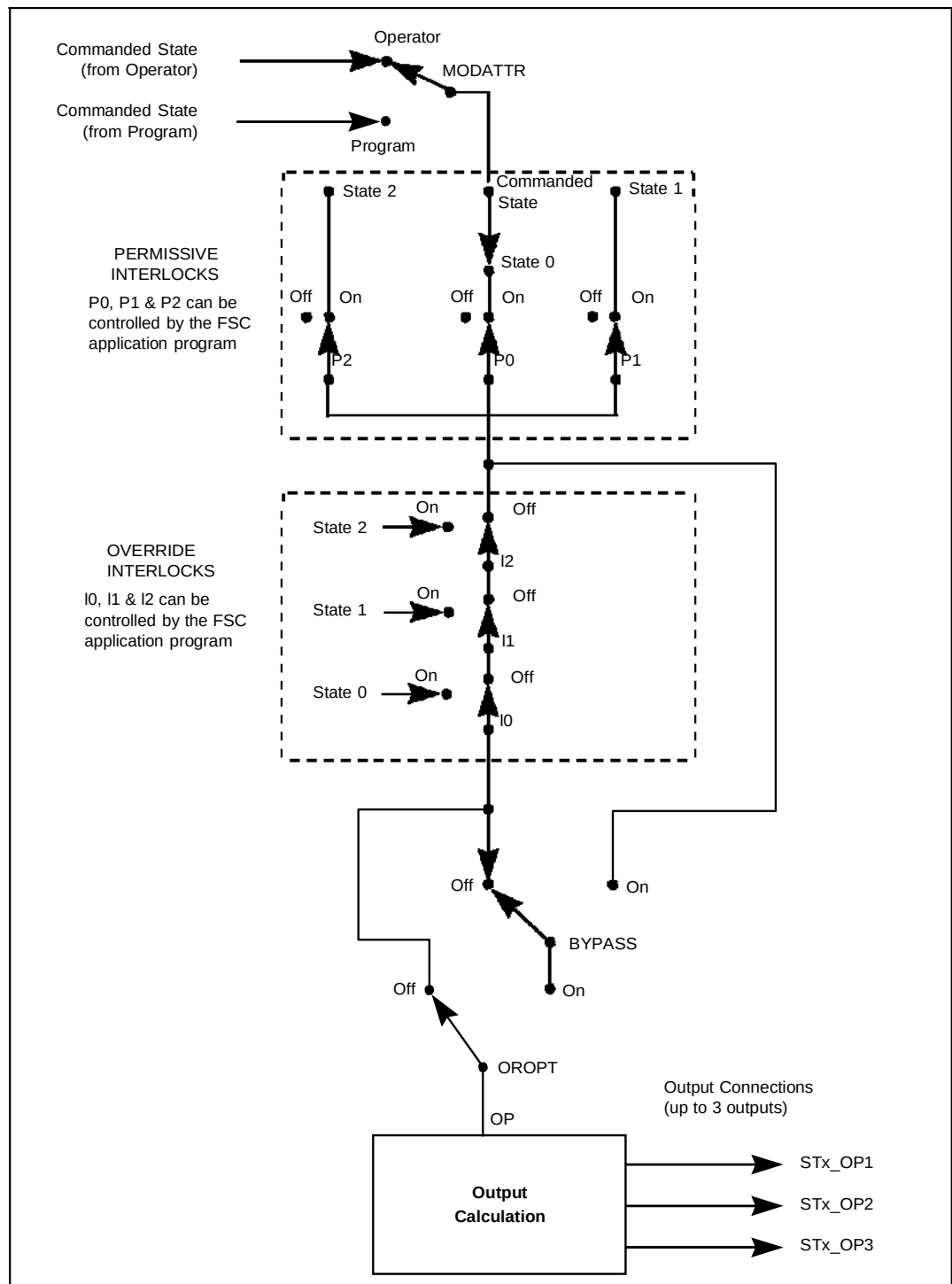


Figure 4-6 – DC Output Processing, Functional Diagram

4.4 Digital Composite Point, Continued

Digital outputs, continued

The three arrows originating from the lower box in Figure 4-6 are not output connections. What the diagram is trying to show is that the DC point uses the configured parameters STx_OPy to determine what values should be written to the output connections.

ATTENTION The following statements apply:

- When override interlocks change, only OP is changed, not the field outputs. (The relation between OP and field outputs is created in the FSC control program.)
- When a commanded state from outside comes in, the override interlock values are checked to verify that the state can be changed. (All override interlocks must be Off.)
- There is no feedback loop from override interlocks to the outputs.

Multiple Output Processing

Parameter NODOPTS can be configured to specify one, two or three FSC-SMM outputs from a DC point, all of which reflect the commanded output state in OP. These outputs can be connected to several destinations, including destinations in the FSC. Where there is more than one FSC output, the SMM compares each read-back output with OP, and reports the first match in OPFINAL. If no match is found, a bad control alarm is generated.

Alarming

A DC point can be configured to detect and report command disagree, uncommanded change, or off-normal alarms through the use of the ALMOPT parameter. Also provided is the option of specifying no alarming for the data point. These alarm options can be configured only if the digital composite point is configured to have inputs, or inputs and outputs.

Command Disagree and Uncommanded Change

This type of alarming is selected by entering Cmddis for the ALMOPT parameter. When the commanded-output state is changed and the actual input PV state does not change accordingly within a predefined feedback time, a command disagree alarm is generated. The feedback time is specified by the FBTIME parameter. This alarm condition returns to normal when the input PV state and the commanded-output state are the same.

Continued on next page

4.4 Digital Composite Point, Continued

Alarming, continued

If the commanded state is momentary, no alarm is generated. For example, a motor may have two PV states (Run and Stop), but there may be three commanded output states (Run, Stop, and Jog) where Jog is defined as a momentary state. Command disagree alarming is performed for only the Run and Stop commanded output states.

If a change does not occur in the commanded output state but the input PV state changes (and the PV is not bad), an uncommanded-change alarm is generated. This alarm condition returns to normal when the input PV state and the commanded state are the same. If the point state has been configured as being momentary, this type of alarm is not applicable.

Off-normal

Detection of off-normal alarms is configured by entering Offnorml for the ALMOPT parameter. The normal state of the PV input is defined by the user through the PVNORMAL parameter. When the PV input state is different from the state specified by the PVNORMAL parameter, the off-normal alarm is generated. The alarm condition returns to normal when the PV input state and the specified PV normal state are the same.

Change-of-State Events

Any transitions in the PV input state can be reported as events for journalizing and for causing the event-initiated processing of points in the Application Module (AM). It is configured by entering EIP for the Event-Report-Option parameter EVTOPT. The user must enter the tag name of the AM or CM point using the EIPPCODE parameter. If only journalizing is required, the EIPPCODE parameter can be set to a null tagname.

Bad Control Alarm

A bad control alarm indicates that the FSC output state does not match the commanded state in the OP parameter. After a change in the output is commanded by an operator at a Universal Station or by a user-written program, the FSC-SMM defers the calculation of the read back status from the FSC for one FSC control program scan to allow the FSC to update its output to the field. The FSC-SMM compares each of the outputs (1, 2 or 3) with OP, and reports the first match in OPFINAL. A Bad Control alarm (BADCTLFL = On) is generated if no match is found on any output. This causes the bad control indicator (see Figure 4-4) to appear on the Group or Detail display for the point.

Continued on next page

4.4 Digital Composite Point, Continued

Local manual indication

Field devices that are interfaced by a DC point often have a local Hand/Off/Auto switch (often called Hand/Off/Remote). Unless this switch is in the Auto position, the SM may not have any control over that device. The user can optionally feedback the switch position into the SM to obtain some display indication for the Universal Station operator. This indication is provided by the word LOCALMAN appearing at the bottom of the DC point on a Group display.

When in local manual, any changes to the output by the operator, user programs or override interlocks are typically prohibited (assuming the FSC is programmed accordingly). The override interlocks should be programmed to take effect as soon as the local manual condition is cleared.

To support Hand/Off/Auto switches, a Boolean flag called LOCALMAN is provided for connection to the FSC. The On state indicates that the switch is not in Auto position.

CAUTION Be aware of the importance of assuring compatibility of the FSC control program with the connections established for the FSC-SMM DC point.

4.5 Analog Input Point

Background

The analog input (AI) point converts raw FSC numbers into engineering units for use by the rest of the TDC 3000 system.

Data flow

Analog input values from the field can be collected from the FSC via its logic control program.

The FSC:

- collects the analog input value from the specified I/O address, and
- stores the collected value in its data storage area (memory), and
- transfers the most recently collected value of the desired analog input from the FSC-to-SMM output table to the FSC-to-SMM input table.

The FSC-SMM:

- accesses the specified location where the desired analog input value has been stored.

Point processing

Upon receiving the raw AI value from the FSC, the FSC-SMM performs:

- PV characterization,
 - range checking and PV filtering,
 - PV source selection, and
 - alarm detection.
-

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4.5 Analog Input Point, Continued

Point processing,
continued

Figure 4-7 illustrates the processing functions and associated parameters.

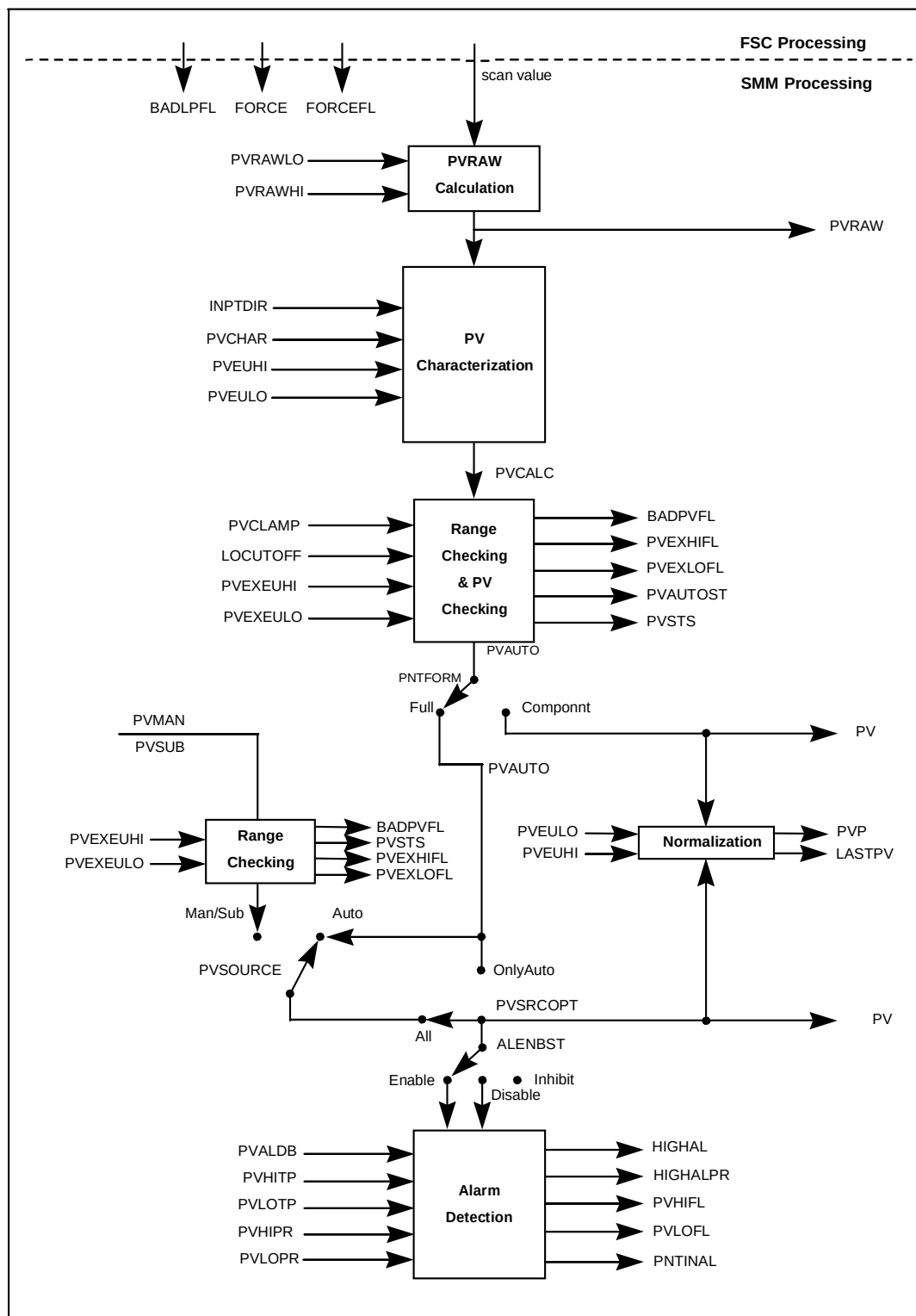


Figure 4-7 – AI Point Processing, Functional Diagram

4.5 Analog Input Point, Continued

Raw PV processing

The value written by the FSC is converted to percent (0 to 100%) using the following equation:

$$PVRAW = 100 * \frac{\text{scan value} - PVRAWLO}{PVRAWHI - PVRAWLO}$$

Maximum PV resolution

The maximum PV resolution is as follows:

$$PV \text{ resolution} = \frac{100}{PVRAWHI - PVRAWLO}$$

ATTENTION This only applies for FSC-SM PI 53.00.
FSC-SM PI 53.01 has no limitations for the PV resolution.

PV characterization

The PVRAW signal received from the FSC is a percentage of the source range. It is characterized based on the entries made for the PVCHAR and INPTDIR parameters. This value is converted to engineering units.

PVCHAR can be Linear or Sqroot. The intermediate variable pvint is calculated as follows:

Table 4-10 – PV Characterization Effect on Characterization Equations

PVCHAR	pvint
Linear	PVRAW
Sqroot	$100 * \sqrt{\frac{PVRAW}{100}}$

The output value of the characterization is PVCALC. The value of the input direction parameter (INPTDIR) is taken into consideration during the calculation of PVCALC as described in Table 4-11, where pvint is the intermediate variable calculated as shown above.

Table 4-11 – Input Direction Effect on Characterization Equations

INPTDIR	PVCALC
Direct	$\left(\frac{pvint}{100} * (PVEUHI - PVEULO) \right) + PVEULO$
Reverse	$PVEUHI - \left(\frac{pvint}{100} * (PVEUHI - PVEULO) \right)$

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4.5 Analog Input Point, Continued

PV range checking

PV range checking ensures that the PVCALC output of the PV characterization is within the limits defined by the parameters PVEXEULO and PVEXEUHI.

PV range checking (limits defined by parameters PVEUHI and PVEULO) for analog input points has the characteristics as listed in Table 4-12.

Table 4-12 – PV range checking

If PVCLAMP is ...	And PVAUTO ...	Then ...	And PV is ...
NoClamp	exceeds PVEUHI	nothing happens.	–
	exceeds PVEXEUHI	PVSTS goes Bad, PVEXHIFL goes On, and BADPVFL goes On.	Set to NaN.
	goes below PVEULO	nothing happens.	–
	goes below PVEXEULO	PVSTS goes Bad, PVEXLOFL goes On, and BADPVFL goes On.	Set to NaN.
Clamp	exceeds PVEUHI	nothing happens.	–
	exceeds PVEXEUHI	PVSTS goes Uncertain, PVEXHIFL goes On, and BADPVFL remains Off.	clamped to PVEXEUHI.
	goes below PVEULO	nothing happens.	–
	goes below PVEXEULO	PVSTS goes Uncertain, PVEXLOFL goes On, and BADPVFL remains Off.	clamped to PVEXEULO.

ATTENTION If the range-checked value is less than the value specified by the user-configured LOCUTOFF parameter, PVAUTO is forced to PVEULO.

PV source selection

The PVSOURCE parameter allows the user to select the source of the PV for this data point. The PV can be provided by the range checking circuit (when PVSOURCE is Auto). In addition, the PV source option parameter (PVSRCOPT) determines whether it is permissible to change the PV source to a source other than Auto. PVSRCOPT has two states: OnlyAuto and All. The All state allows the PVSOURCE to be set to Man (operator can enter value) or Sub (program can enter value).

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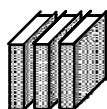
4.5 Analog Input Point, Continued

Alarming

The analog input data point compares the PV to threshold values and records the alarms in the database of the data point. The alarms are then reported by the FSC-SMM. Table 4-13 lists the parameters that are associated with alarming in the analog input point.

Table 4-13 – Analog Input Point Associated Alarm Parameters

Parameter	Definition
ALENBST	Alarm Enable Status
BADPVFL	Bad PV Flag
BADPVPR	Bad PV Alarm Priority
CONTCUT	Contact Cutout
EIPPCODE	Event-Initiated Processing Point Identifier
HIGHAL	Highest Alarm Detected
HIGHALPR	Highest Level Alarm's Priority
PTINAL	Point in Alarm Indicator
PVALDB	PV Alarm Deadband as a Percent of Full Range
PVEXHIFL	PV Extended High Range Violation
PVEXLOFL	PV Extended Low Range Violation
PVHIFL	PV High Alarm Flag
PVHIPR	PV High Alarm Priority
PVHITP	PV High Alarm Trip Point
PVLOFL	PV Low Alarm Flag
PVLOPR	PV Low Alarm Priority
PVLOTP	PV Low Alarm Trip Point



Refer to the *FSC Safety Manager Parameter Reference Dictionary* for detailed definitions of these parameters.

4.6 Analog Output Point

Background

The analog output (AO) point value is copied by the FSC from the SMM-to-FSC output table to the SMM-to-FSC input table.

Point processing

Figure 4-8 illustrates the FSC-SMM processing functions that are associated with an analog output point, and can be configured via parameters.

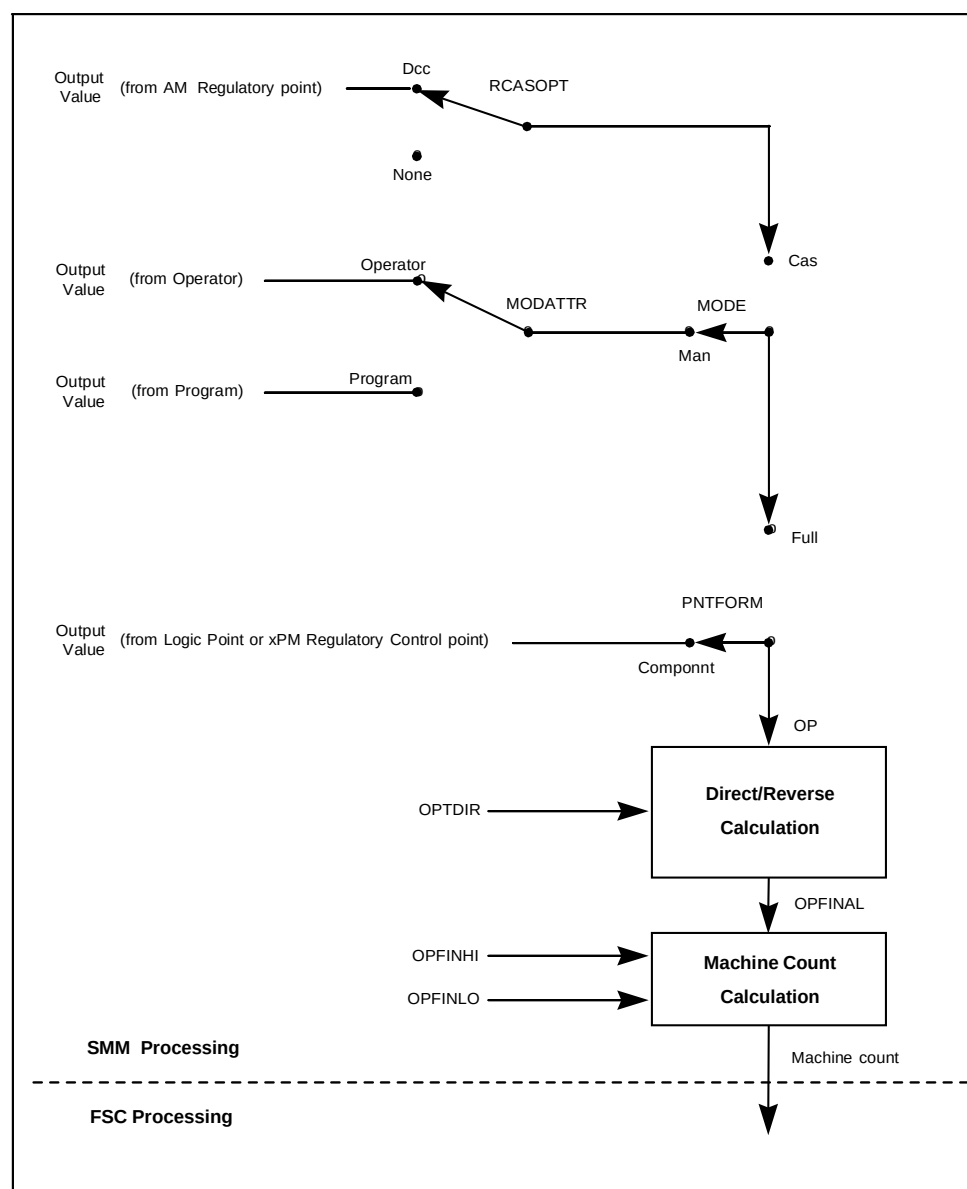


Figure 4-8 – AO Point Processing, Functional Diagram

Continued on next page

4.6 Analog Output Point, Continued

Output direction

The OPTDIR parameter specifies whether the output of the data point is direct or reverse directed:

Table 4-14 – Output Direct Effect

OPTDIR	OPFINAL
Direct	OP
Reverse	100 – OP

Final value processing

The value read by the FSC is calculated using the following equation:

$$\text{machine count} = \left(\frac{\text{OPFINAL}}{100} * (\text{OPFINHI} - \text{OPFINLO}) \right) + \text{OPFINLO}$$

4.7 Logic Point

Background

The function of the logic point is limited to the transfer of data between points in modules on the same UCN. The logic point has, therefore, in the LM been referred to as the "linkage point", and is the basis for the FSC-SMM's application-level support of peer-to-peer communications. Data types can be Boolean or Real.

Configuration

Each logic point accommodates 12 input connections and 12 output connections, with a total maximum of 50 connections scheduled for processing during any single FSC-SMM scan cycle.

Input connections

Inputs to the logic point are assigned during configuration by using logic input connections. Up to 12 inputs can be assigned to logic-slot inputs L(1) to L(12), as shown in Figure 4-9. The logic input connections are specified through the LISRC parameter.

The inputs to the logic slot can be obtained from:

- any Boolean, integer, unsigned integer, enumeration, self-defining enumeration, or real parameter within this FSC-SMM, or in another node on the same UCN, using the "Tagname.Parameter" format.

ATTENTION

Integer, unsigned integer, enumeration or self-defining enumeration parameters are automatically converted to real values.

- an address (!LCxxxxx) in the FSC.

ATTENTION

Logic point input and output must be of the same type, i.e. you can connect Flag1.PV to a Flag2.PV (enum ☒ enum), but not a Flag1.PV to a Flag2.PVFL (enum ☒ boolean).

Continued on next page

4.7 Logic Point, Continued

Input connections, continued

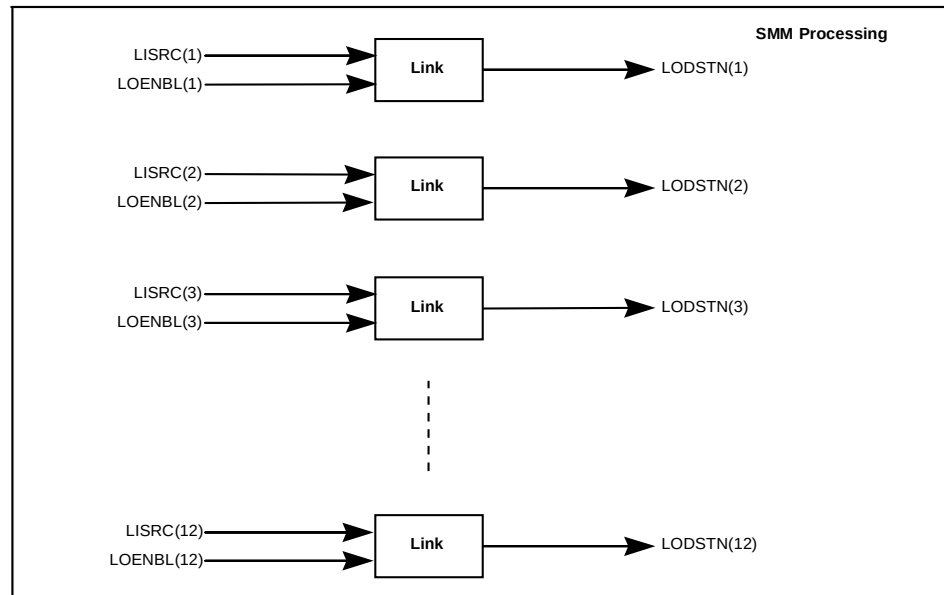


Figure 4-9 – Logic Point Input Connections

Input errors

Three types of errors can occur with regard to logic point inputs.

- A communication error occurs as the result of a failed device.
- The point has been deleted.
- A configuration error occurs when the point mix in that module has been changed such that the specified point no longer exists.

In order for the routine to be able to continue in spite of a configuration error, the following special features are provided:

- Bad Boolean inputs - If a Boolean input is not successfully fetched, its value is defaulted, based on the logic-input-bad-handling-option parameter (LIBADOPT) as listed in Table 4-15.
- Bad real inputs - If a real input cannot be successfully fetched, its value is defaulted to NaN (Not a Number) and no value is set to the output. This means that the destination keeps its last good value.

Table 4-15 – Logic Input Point Bad Handling Option Parameters

LIBADOPT	Action
On	The On state is substituted for the unsuccessful input.
Off	The Off state is substituted for the unsuccessful input.
Hold	The previous value (the last successfully fetched value) is substituted for the unsuccessful input. On start-up, the previous value is defaulted to the Off state.

Continued on next page

4.7 Logic Point, Continued

Output connections

Logic output connections are used to write the values of local parameters of a logic slot to the configured destinations. Up to 12 output connections can be configured for each logic slot. The destinations are specified by the LODSTN parameter by using the "Tagname.Parameter" format, the UCN hardware reference address format, or the FSC address. Each destination address is tied to the input bearing the same number (LODSTN(1) to L1, LODSTN(2) to L2, etc.).

The logic output connection can write the selected local parameters of a logic slot to

- any Boolean or real parameter in this FSC-SMM, or another node on the same UCN,
- an address in the FSC of which the corresponding variable has an DCS address.

Associated with each output connection is a logic output enable flag, LOENBL(n), which is always an FSC address (!LCxxxxx). The FSC variable pointed to by LOENBL(n) must be On for the corresponding output connection to write to the specified destination. The default value of 0 represents the On state.

Generic descriptors

Up to twelve user-defined generic descriptors are provided with each logic slot for identifying external inputs (L1, L2, etc.) with custom names to be shown on Universal Station displays. The actual number of descriptors used is determined by the NODESC parameter for this logic slot. The corresponding eight-character descriptor is defined by GENDESC(n).

4.8 Flag Point

Background

A flag data point is a two-state (On and Off) point that is used for storing a Boolean value.

Data flow

A Flag is not just a Boolean value in the FSC-SMM. The value of each Flag is copied to the FSC every FSC cycle, using the mechanism illustrated in Figure 2-2.

The Flag point's state can change whenever it is accessed by other system functions, such as an operator, a continuous point or a user-written program.

The point's state can be supplied by

- the operator,
- an output connection from another FSC-SMM point (DC or LS),
- another box (SM, PM, LM, etc.) on the same UCN,
- a node on the LCN (AM etc.).

The FSC cannot write the value of the flag.

Point processing

The flag point has the parameters PVFL and PV. PVFL has a Boolean value, whereas PV has the value equal to either STATETXT(0) or STATETXT(1).

PVFL will also light the Pvstate1 and Pvstate0 boxes on the Universal Station displays, depending on its state. If PVFL is On, the Pvstate1 (upper) box will be lit; if PVFL is Off, the Pvstate0 (lower) box will be lit.

Flag points can be configured as full or component points as determined through the PNTFORM parameter.

Figure 4-10 illustrates the FSC-SMM processing associated with the flag point.

Continued on next page

4.8 Flag Point, Continued

Point processing, continued

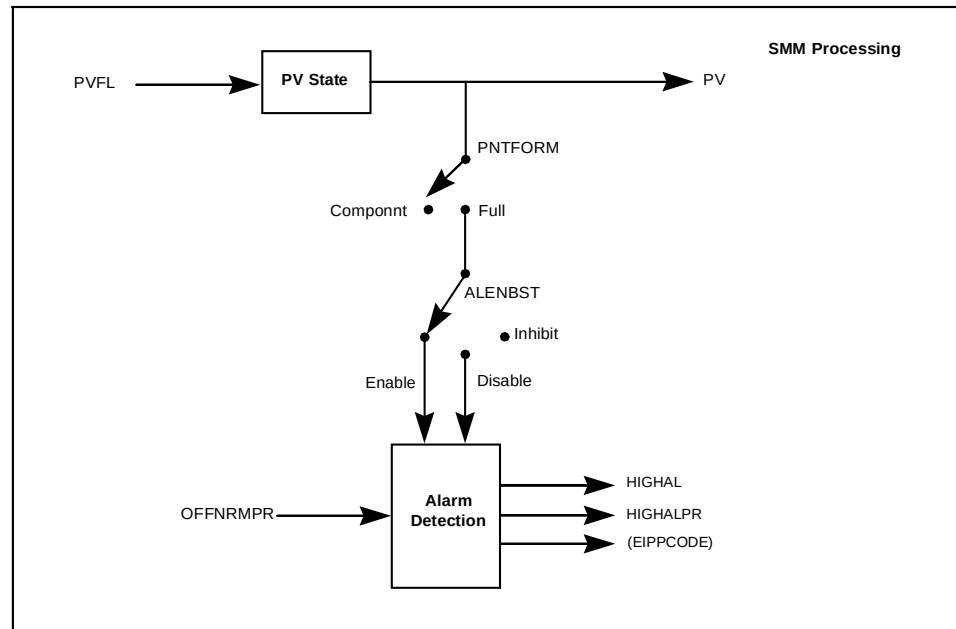


Figure 4-10 – Flag Point Processing, Functional Diagram

Alarming

Flags with $\text{SLOTNUM} \leq 512$ can be configured for off-normal alarming. An alarm is generated when the PV of the flag point is changed from Pvstate0 (PVFL = Off) to Pvstate1 (PVFL = On). Alarming is available only if the flag point has been configured as a full point.

4.9 Numeric Point

Background

Numeric data points are used to store real values, such as

- operator entries,
- accumulations, or
- intermediate results of program calculations.

Data flow

A Numeric is not just a real value in the FSC-SMM. The value of each Numeric is copied to the FSC every FSC cycle, using the mechanism illustrated in Figure 2-2.

The Numeric point's state can change whenever it is accessed by other system functions, such as an operator, a continuous point, or a user-written program.

The point's value can be supplied by:

- the operator,
 - an output connection from another FSC-SMM point (LS),
 - another box (SM, PM, LM, etc.) on the same UCN,
 - a node on the LCN (AM etc.).
-

4.10 Timer Point

Background

The timer data point provides timekeeping in the FSC Safety Manager. This type of data point keeps track of the elapsed time after the timer has been started (by the operator or by a program), and provides an indication when the elapsed time has reached the predefined limit.

Point processing

Timer points can be configured as either full or component points. They are accessible to routines in:

- the same SM,
- any other box (SM, PM, APM, etc.) on the same UCN,
- the FSC, or
- any node on the LCN.

The preset length of time that the timer point is to run – in units of tenths of a second, seconds or minutes – is determined by the value recorded in the SP parameter. The SP parameter is set to the preset time value by the operator at a Universal Station, or by a program.

This value is frozen for one FSC-SMM cycle (0.5 second) to assure transfer to the FSC. Once the preset value is entered, the timer can be started by setting the COMMAND parameter to Start. The operator has also the option to reset the timer through the COMMAND parameter.

The time value in PV starts at 0 and increments toward the preset time value. The RV parameter indicates the time remaining until the timer reaches its limit (SP - PV). When PV = SP, the parameter SO is set to On indicating the timer has elapsed. The values in PV and SP can range from 0 to 2047 timebase units, in accordance with the FSC program. Thus the maximum time a timer can run is 34 hours and 7 minutes. The value of PV is an integer value. Thus, for example, if the TIMEBASE is minutes, the PV is updated every minute, and remains constant during one minute.

The timer point in an FSC-SMM does not perform the timer function itself. This is done by a Timer variable in the control program in the FSC. This FSC timer is a countdown timer, and communicates the RV to the FSC-SMM. The FSC-SMM Timer point translates the COMMAND parameter to the Boolean signals Set and Reset, and calculates PV from SP and RV.

Continued on next page

4.10 Timer Point, Continued

Point processing, continued

If the command Start is given when the timer is in STATE Disable, the timer starts running, and the PV increases from 0. If the command Start is given when the timer is in STATE Enable, the value of PV is set to 0, and the timer keeps running.

When the command Reset is given when the timer is in STATE Disable, the timer remains stopped, and the PV remains 0. When the command Reset is given when the timer is in STATE Enable, the timer stops running, and the PV is set to 0.

It is not possible to write SP with time running. Also, if the timer has stopped and SP is written to another larger value, the timer will begin timing up to the new SP value.

Figure 4-11 illustrates the FSC-SMM processing associated with the timer point.

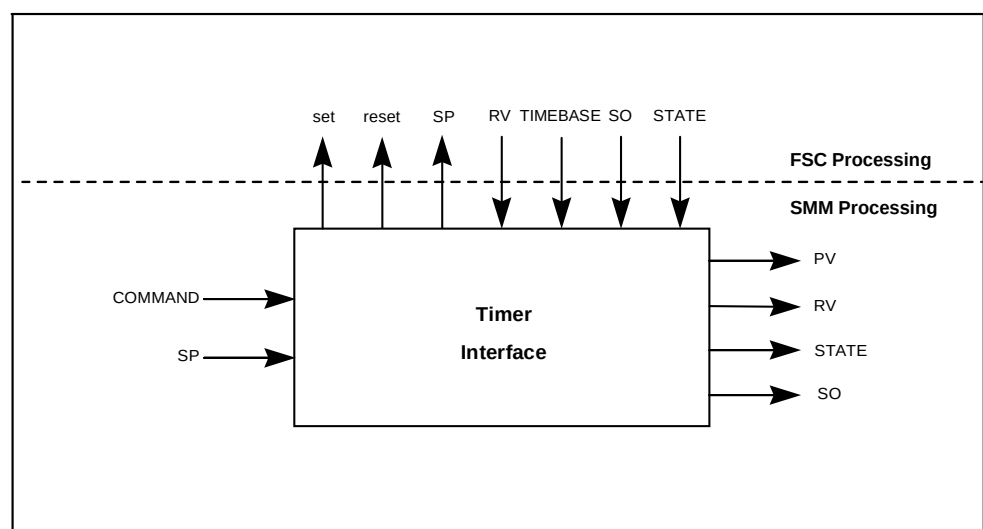


Figure 4-11 – Timer Point Processing, Functional Diagram

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